DGA-MUON: DECOUPLED GEOMETRY-ALIGNED ADAPTIVE SCALING FOR MUON

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ABSTRACT

While NorMuon has achieved strong empirical performance in pretraining, its underlying adaptive mechanism remains largely heuristic and poorly understood. In this work, we provide the first systematic theoretical analysis of NorMuon's adaptivity, revealing that it primarily arises from orthogonalization-induced geometry rather than genuine optimization dynamics, serving to offset the resulting geometric non-uniformity. Under exact orthogonalization, the adaptive scaling factors degenerate into a single global scalar for square and wide matrices, while for tall matrices their variation results from unevenly distributed row energy after orthogonalization. Under approximate orthogonalization, the orthogonality residuals introduce additional variation into the scaling, giving rise to a counter-intuitive Orthogonalization–Adaptivity Paradox: more accurate orthogonalization weakens adaptivity. We further show that NorMuon's rigid row-wise scaling is geometrically misaligned with the one-sided orthogonal structure of tall matrices by distorting column orthogonality. Motivated by these limitations, we propose two core design principles that a desirable adaptive mechanism for Muon should satisfy. First, adaptive scaling should be decoupled from orthogonalization, with the scaling factors computed directly from raw gradients. Second, adaptive scaling should be aligned with the shape-dependent orthogonal structure of the polar factor, using row-wise scaling for wide matrices and column-wise scaling for tall matrices. We prove that this geometry-aligned scaling preserves the orthogonal structure of the update. By incorporating several other techniques, we obtain Decoupled Geometry-Aligned Muon (DGA-Muon). We establish convergence guarantees for DGA-Muon and empirically validate both our theoretical characterization of NorMuon's scaling degeneration and the superiority of DGA-Muon.

1 INTRODUCTION

The Muon (MomentUm Orthogonalized by Newton–Schulz) optimizer (Jordan, 2024), specifically designed for optimizing two-dimensional parameters in neural networks, first gained traction through its remarkable empirical success in the Modded-NanoGPT speedrun (Jordan et al., 2024), where it demonstrated substantially faster convergence than AdamW (Loshchilov & Hutter, 2019). Unlike AdamW, which treats weight matrices as flattened one-dimensional vectors, Muon preserves their inherent two-dimensional structure and applies an orthogonalization operation to equalize their singular values. Following this initial success, subsequent studies (Liu et al., 2025; Shah et al., 2025) further validated Muon's advantages over AdamW in the pretraining of multi-billion-parameter language models. To date, Muon has been widely adopted for training frontier foundation models, including open-weight models such as Kimi (Team et al., 2025; 2026), GLM (Zeng et al., 2025; 2026), and DeepSeek (Xu et al., 2026a).

The remarkable success of Muon has sparked a surge of research interest. One line of research has sought to understand the mechanisms underlying Muon's superior performance (Nguyen et al., 2026; Wang et al., 2026a;b) and to establish theoretical guarantees for its convergence (Li & Hong, 2025; Kim & Oh, 2026). An even larger body of research is devoted to further enhancing the practical performance of Muon. One prominent branch of these efforts focuses on developing new methods that improve the accuracy or efficiency of the orthogonalization procedure (Amsel et al., 2026; Zhang et al., 2026a; Liu et al., 2026). Another promising direction explores adaptive preconditioning for

DGA-MUON: 解耦几何对齐自适应缩放公式

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摘要

虽然NorMuon在预训练中取得了强大的经验性能，但其底层的自适应机制仍然很大程度上是启发式的，理解也不够深入。在这项工作中，我们首次对NorMuon的自适应进行了系统的理论分析，揭示其主要源于正交化引起的几何结构，而非真实的优化动力学，目的是抵消由此产生的几何非均匀性。在精确正交化的情况下，自适应缩放因子对于方阵和宽矩阵退化为一个全局标量，而对于高矩阵，其变化源于正交化后行能量的不均匀分布。在近似正交化的情况下，正交残差为缩放引入了额外的变化，产生了一个反直觉的正交化-适应性悖论：更精确的正交化会削弱自适应能力。我们进一步证明，NorMuon的刚性行缩放与高矩阵的单边正交结构在几何上是不对齐的，因为它扭曲了列正交性。受这些限制的启发，我们提出了两个核心设计原则，一个理想的Muon自适应机制应该满足。首先，自适应缩放应该与正交化解耦，缩放因子直接从原始梯度计算。其次，自适应缩放应该与极因子的形状相关正交结构对齐，对宽矩阵使用行缩放，对高矩阵使用列缩放。我们证明这种几何对齐的缩放保留了更新的正交结构。通过结合其他几种技术，我们获得了解耦几何对齐Muon（DGA-Muon）。我们为DGA-Muon建立了收敛保证，并经验验证了NorMuon缩放退化的理论特征以及DGA-Muon的优越性。

1 简介

μ子（由牛顿-舒尔茨正交化的动量）优化器（Jordan, 2024），专门设计用于优化神经网络中的二维参数，最初通过在Modded-NanoGPT加速跑（Jordan等人, 2024）中的显著实证成功而获得关注，

在那里它比AdamW（Loshchilov & Hutter, 2019）表现出显著更快的收敛速度。与AdamW不同，后者将权重矩阵视为展平的一维向量，μ子保留了它们固有的二维结构，并对它们的奇异值进行正交化操作以实现均衡。在此初步成功之后，后续研究（Li等人, 2025; Shah等人, 2025）进一步验证了μ子在数十亿参数语言模型的预训练中相对于AdamW的优势。迄今为止，μ子已被广泛用于训练前沿基础模型，包括Kimi（团队等人, 2025; 2026）、GLM（曾等人, 2025; 2026）和DeepSeek（徐等人, 2026a）等开放权重模型。

Muon的显著成功激发了大量研究兴趣。一条研究路线致力于理解Muon卓越性能背后的机制（Nguyen等人，2026；Wang等人，2026a;b）并为其收敛性建立理论保证（Li & Hong, 2025；Kim & Oh，2026）。更大规模的研究致力于进一步提升Muon的实际性能。这些努力中一个突出的分支专注于开发改进正交化过程精度或效率的新方法（Amsel等人，2026；Zhang等人，2026a；Liu等人，2026）。另一个有前景的方向探索自适应预条件化

Muon to better account for the geometry of the loss landscape. Conceptually, this mirrors how adaptive optimizers such as AdaGrad (Duchi et al., 2011) and Adam (Kingma & Ba, 2015) improved upon vanilla SGD. More fundamentally, a theoretical study (Su, 2025) suggests that the perfectly uniform spectral flattening induced by Muon may not be optimal, providing a principled foundation for incorporating adaptive preconditioning into Muon. While several initial attempts have been made, most of these methods either incur prohibitive computational or memory costs (e.g., requiring expensive matrix decompositions or maintaining full-matrix preconditioners) (Xu et al., 2026b; Zhang et al., 2026c; Song et al., 2026) or degrade to a simplistic single global scaling factor (Zhang et al., 2025a). Striking an effective balance between computational efficiency and adaptive expressiveness, NorMuon (Li et al., 2026) has emerged as the most widely adopted adaptive Muon variant. By employing a lightweight yet fine-grained neuron-wise (i.e., row-wise) scaling strategy, NorMuon consistently outperforms vanilla Muon across various settings. Notably, it achieves state-of-the-art performance on two leading speedruns: Modded-NanoGPT and NanoChat (Karpathy, 2025).

A distinctive characteristic of NorMuon is that its adaptive scaling factors are computed from the orthogonalized momentum, rather than directly from historical gradients as in conventional adaptive optimizers such as AdaGrad (Duchi et al., 2011), Adam (Kingma & Ba, 2015), and Shampoo (Gupta et al., 2018). This design tightly couples adaptivity with the orthogonalization procedure, constituting a fundamentally different paradigm for adaptive optimization. However, this design choice is largely heuristic, and the rationale for using orthogonalized momentum as the source of adaptivity lacks theoretical justification. Thus far, neither theoretical convergence guarantees for NorMuon nor a principled characterization of its adaptive behavior have been established, leaving this unconventional adaptive mechanism poorly understood.

In this work, we provide the first systematic theoretical analysis of NorMuon's adaptivity. To understand its nature, we analyze the variation of NorMuon's adaptive scaling factors, asking what information is encoded in their differences and what drives this variation. Surprisingly, we find that NorMuon's adaptivity does not originate from conventional optimization dynamics, but instead arises primarily from the geometry induced by orthogonalization. To see this clearly, recall that NorMuon's scaling factors are derived from the squared ℓ_2-norms of the rows of the orthogonalized matrix. Although orthogonalization equalizes singular values, it does not necessarily imply uniform row-wise norms. In fact, under exact orthogonalization, this row-wise non-uniformity constitutes the primary source of variation in the scaling factors. However, such variation arises exclusively for tall matrices, where orthogonalization enforces only one-sided orthogonality, namely column orthonormality, but not row orthonormality. Consequently, while the total squared ℓ_2-norm across all rows is fixed, the squared ℓ_2-norm of each individual row remains unconstrained, allowing the row-wise scaling factors to vary (see Subsection 3.3 for a full derivation). In contrast, square and wide matrices satisfy row orthonormality, which ensures that the squared ℓ_2-norm of every row is exactly one. Consequently, the row-wise adaptive scaling factors become identical across rows, and the adaptive scaling completely degenerates into a single global factor, thereby nullifying the intended row-wise adaptivity (see Subsection 3.1 for detailed analysis). In practical implementations, orthogonalization is typically approximate. The resulting orthogonality residuals introduce an additional source of variation into the scaling factors. Notably, this gives rise to a rather counterintuitive and undesirable phenomenon, which we term the Orthogonalization–Adaptivity Paradox: the more accurate the orthogonalization, the weaker the resulting adaptivity, which vanishes entirely under exact orthogonalization for square and wide matrices; conversely, less accurate orthogonalization permits greater variation in the scaling factors and thus stronger adaptivity (see Subsections 3.2 and 3.3 for details). Together, these observations reveal a fundamental limitation of NorMuon: its adaptive scaling primarily acts as a compensatory mechanism that offsets geometric non-uniformity induced by orthogonalization, rather than capturing genuine optimization-relevant information.

The preceding analysis of NorMuon's inherent limitation suggests that a desirable adaptive mechanism for Muon should preserve NorMuon's computational efficiency while deriving adaptivity from optimization-relevant signals rather than orthogonalization-induced geometry. At its core, NorMuon's limitation arises from computing scaling factors from the orthogonalized momentum, which inevitably causes the resulting adaptivity to inherit geometric distortions induced by orthogonalization. To break this coupling, we introduce a decoupled design principle: computing adaptive scaling factors directly from raw gradients, independent of orthogonalization, and subsequently applying them to the orthogonalized matrix, i.e., the polar factor (Higham, 2008; 2020) of the momentum. Another design deficiency of NorMuon lies in its reliance on rigid row-wise scaling, which overlooks

Muon的自适应预条件化以更好地考虑损失函数的几何结构。从概念上讲，这类似于AdaGrad（Duchi等人，2011）和Adam（Kingma & Ba，2015）等自适应优化器如何改进基础随机梯度下降。更根本地，一项理论研究（Su，2025）表明，Muon引起的完美均匀频谱展平可能并非最优，为将自适应预条件化纳入Muon提供了基本原则。尽管已经进行了一些初步尝试，但这些方法中的大多数要么带来过高的计算或内存成本（例如，需要昂贵的矩阵分解或维护完整矩阵预条件器）（Xu等人，2026b；Zhang等人，2026c；Song等人，2026），要么退化为简单的单一全局缩放因子（Zhang等人，2025a）。在计算效率和自适应表达能力之间取得有效平衡，NorMuon（Li等人，2026）已成为最广泛采用的自适应Muon变体。通过采用轻量级但细粒度的逐神经元（即逐行）缩放策略，NorMuon在各种设置中始终优于基础Muon。值得注意的是，它在两个领先的速度测试中实现了最先进的性能：Modded-NanoGPT和NanoChat（Karpathy，2025）。

NorMuon 的一个显著特征是，其自适应缩放因子是从正交化动量计算得到的，而不是像传统自适应优化器（如 AdaGrad (Duchi 等人，2011 年)、Adam (Kingma & Ba，2015 年) 和 Shampoo (Gupta 等人，2018 年)）那样直接从历史梯度计算。这种设计将自适应性与正交化过程紧密耦合，构成了自适应优化的根本不同范式。然而，这种设计选择在很大程度上是启发式的，使用正交化动量作为自适应性的来源缺乏理论依据。迄今为止，NorMuon 的理论收敛保证及其自适应行为的原则性特征均未建立，导致这种非常规的自适应机制理解不足。

在这项工作中，我们首次对 NorMuon 的自适应性进行了系统的理论分析。为了理解其本质，我们分析了 NorMuon 自适应缩放因子的变化，探究其差异中编码了哪些信息以及是什么驱动了这种变化。令人惊讶的是，我们发现 NorMuon 的自适应性并非源于传统优化动力学，而是主要源于正交化引起的几何形状。要清晰地看到这一点，请回忆

NorMuon 的缩放因子是从正交化矩阵的行向量的平方 ℓ_2-范数推导出来的。尽管正交化使奇异值相等，但这并不一定意味着行范数均匀。事实上，在精确正交化的情况下，这种行非均匀性构成了缩放因子变化的主要来源。然而，这种变化仅出现在高维矩阵中，其中正交化强制执行单边正交性，即列正交规范性，但不是行正交规范性。因此，虽然所有行的平方 ℓ_2-范数的总和是固定的，但每个单独行的平方 ℓ_2-范数仍然不受约束，允许行自适应缩放因子变化（有关完整推导，请参见第 3.3 节）。相比之下，方阵和宽矩阵满足行正交规范性，这确保了每一行的平方 ℓ_2-范数正好为 1。因此，行自适应缩放因子在所有行中变得相同，自适应缩放完全退化为单个全局因子，从而消除了预期的行自适应性（有关详细分析，请参见第 3.1 节）。在实际实现中，正交化通常是近似的。由此产生的正交残差为缩放因子引入了另一个变化来源。值得注意的是，这导致了一种相当反直觉且不受欢迎的现象，我们称之为正交化-适应性悖论：正交化越准确，结果的自适应性越弱，在方阵和宽矩阵的精确正交化下完全消失；相反，不太精确的正交化允许缩放因子有更大的变化，从而更强的自适应性（有关详细信息，请参见第 3.2 节和第 3.3 节）。综合这些观察结果，揭示了 NorMuon 的一个基本限制：其自适应缩放主要充当补偿机制，抵消正交化引起的几何非均匀性，而不是捕捉真正的优化相关信息。

前面对NorMuon固有局限性的分析表明，对于Muon而言，一个理想的自适应机制应当保留NorMuon的计算效率，同时从与优化相关的信号而非正交化引起的几何中获取自适应性。其核心在于，NorMuon的局限性源于从正交化动量中计算缩放因子，这不可避免地导致最终的自适应性继承了正交化引起的几何畸变。为了打破这种耦合，我们引入了一种解耦设计原则：直接从原始梯度计算自适应缩放因子，独立于正交化，然后将其应用于正交化矩阵，即动量的极因子（Higham, 2008; 2020）。NorMuon的另一个设计缺陷在于其依赖于僵化的行缩放，这忽略了

a fundamental geometric reality: for non-square matrices, the polar factor inherently exhibits only one-sided orthogonality. Specifically, for tall matrices, the polar factor is column-orthogonal, with its columns defining the orthogonal directions. Row-wise scaling, however, is a one-sided operation applied exclusively along the row dimension, which is misaligned with the column-wise orthogonal structure. As a result, this geometric mismatch distorts the column geometry and breaks the original orthogonality induced by the polar factor. To prevent this structural distortion, we introduce our second design principle: geometry-aligned scaling, which aligns the scaling with the one-sided orthogonal structure. Specifically, the scaling axis is chosen according to the shape-dependent orthogonality structure of the polar factor: row-wise scaling is applied for wide matrices, whereas column-wise scaling is applied for tall matrices (or either for square matrices), thereby preserving the orthogonal structure obtained through orthogonalization. We further provide a formal theoretical justification for the orthogonality-preserving property of the proposed adaptive scaling mechanism (see Subsection 4.2 for the proof).

Beyond the two core design principles, we incorporate several additional design considerations to better regulate the dynamics of the adaptive scaling factors. First, we construct the second-moment estimate for each adaptive unit (i.e., row or column) from the sum, rather than the mean, of the squared gradient entries, thereby making the scaling of each unit depend on its total accumulated gradient energy. This differs from NorMuon, which normalizes the total row-wise gradient energy by the row dimension, making its adaptive scaling depend on average per-coordinate energy. Second, we apply standard bias correction to the second-moment estimates to mitigate the initialization bias of the exponential moving average (EMA) during the early stages of optimization. Third, to curb the excessive anisotropy that adaptive scaling can introduce across different orthogonal directions, we employ an adaptive clipping mechanism that bounds the dynamic range of the scaling factors. This preserves Muon's favorable ability to support large step sizes while maintaining training stability. Combining all these ingredients, we propose Decoupled Geometry-Aligned Muon (DGA-Muon; see Algorithm 1), which equips Muon with geometry-aligned adaptive scaling based on genuine gradient signals while achieving computational efficiency comparable to that of NorMuon.

We establish convergence guarantees for DGA-Muon under standard assumptions in the non-convex setting. We empirically verify our theoretical characterization of NorMuon, showing that removing its adaptive scaling on the square Q/K/V/O projection matrices in NanoChat leaves the validation performance essentially unchanged. We further demonstrate that DGA-Muon substantially outperforms NorMuon in an end-to-end NanoChat evaluation, validating the effectiveness of our proposed adaptive scaling mechanism. Finally, we note that our analysis and algorithmic design provide broader insights into general matrix-aware optimizer design beyond enhancing Muon or NorMuon. Our proposed principles extend naturally to directions other than those used by Muon. More generally, any directions that meaningfully characterize the underlying matrix can serve as candidates for adaptive scaling. Furthermore, our analysis highlights the intrinsic necessity of row-wise/column-wise adaptivity as a natural mechanism for adapting to the typical one-sided directional structure inherent in matrices, which manifests along either the row or column dimension.

2 Preliminaries: Muon and NorMuon

In this section, we briefly review Muon and NorMuon, providing the basis for our subsequent theoretical analysis.

Muon. Muon is an optimizer tailored for two-dimensional matrix parameters. It maintains a momentum buffer M_t, updated according to $M_t = \beta_1 M_{t-1} + (1-\beta_1)G_t$, where G_t denotes the gradient at iteration t, $\beta_1 \in [0,1)$ is the momentum coefficient, and $M_0 = \mathbf{0} \in \mathbb{R}^{m\times n}$. However, rather than updating the parameters directly along the raw momentum direction, Muon applies the update along the direction of the orthogonalized momentum. Specifically, given the current weight matrix $W_{t-1} \in \mathbb{R}^{m\times n}$ and the momentum $M_t \in \mathbb{R}^{m\times n}$, the update rule of Muon is given as follows

$$W_t = W_{t-1} - \eta\,\mathrm{Orth}(M_t)$$

where η is the learning rate, and $\mathrm{Orth}(M_t)$ denotes the orthogonalization of M_t, i.e., $\mathrm{Orth}(M_t) = UV^\top$ with $M_t = U\Sigma V^\top$ being the (thin) singular value decomposition (SVD). The orthogonalization procedure discards all singular value information by setting every singular value to one, while preserving the directional structure induced by the singular vectors. The resulting orthogonalized

一个基本的几何事实：对于非方阵，极因子本质上只表现出单侧正交性。具体来说，对于高矩阵，极因子是列正交的，其列定义了正交方向。然而，行缩放是一种仅沿行维度应用的单侧操作，这与列正交结构不一致。因此，这种几何不匹配扭曲了列几何，并破坏了极因子引起的原始正交性。为了防止这种结构畸变，我们引入了我们的第二个设计原则：几何对齐缩放，该原则将缩放与单侧正交结构对齐。具体来说，缩放轴根据极因子的形状依赖的正交结构选择：对于宽矩阵应用行缩放，而对于高矩阵（或方阵）应用列缩放，从而保留通过正交化获得的正交结构。我们进一步为所提出的自适应缩放机制的正交性保持特性提供了形式化的理论证明（证明见第4.2节）。

在两个核心设计原则之外，我们结合了若干附加设计考量，以更好地调控自适应缩放因子的动态特性。首先，我们基于平方梯度项的总和而非均值来构建每个自适应单元（即行或列）的二阶矩估计，从而使每个单元的缩放取决于其累积的梯度能量总和。这与NorMuon不同，后者通过行维度对总行向梯度能量进行归一化，使其自适应缩放依赖于每个坐标的平均能量。其次，我们对二阶矩估计应用标准偏差校正，以减轻指数移动平均（EMA）在优化早期阶段的初始化偏差。第三，为了抑制自适应缩放可能在不同正交方向上引入的过度各向异性，我们采用自适应裁剪机制来界定缩放因子的动态范围。这既保留了Muon支持较大步长的有利能力，又维持了训练稳定性。结合所有这些要素，我们提出了解耦几何对齐自适应缩放公式（DGA-Muon；参见算法1），该公式使Muon具备基于真实梯度信号的正交对齐自适应缩放能力，同时实现了与NorMuon相当的计算效率。

我们在非凸设置下建立了 DGA-Muon 的收敛保证。我们通过实验验证了 NorMuon 的理论特性，表明在 NanoChat 中移除其在正方形 Q/K/V/O 投影矩阵上的自适应缩放后，验证性能基本保持不变。我们进一步证明，在端到端的 NanoChat 评估中，DGA-Muon 的性能显著优于 NorMuon，验证了我们提出的自适应缩放机制的有效性。最后，我们指出，我们的分析和算法设计为更广泛的矩阵感知优化器设计提供了更深入的见解，而不仅仅是增强 Muon 或 NorMuon。我们提出的原理自然地扩展到 Muon 未使用的方向。更普遍地说，任何能够有意义地表征底层矩阵的方向都可以作为自适应缩放的候选。此外，我们的分析突出了行/列自适应的内在必要性，作为适应矩阵中典型单边方向结构（该结构沿行或列维度显现）的自然机制。

2 预备知识：Muon 与 NorMuon

在本节中，我们简要回顾了Muon和NorMuon，为后续的理论分析奠定了基础。

Muon。 Muon是为二维矩阵参数设计的优化器。它维护一个动量缓存 M_t，根据 $M_t = \beta_1 M_{t-1} + (1-\beta_1)G_t$进行更新，其中 G_t 表示迭代 t处的梯度， $\beta_1 \in [0,1)$是动量系数，以及 $M_0 = \mathbf{0} \in \mathbb{R}^{m\times n}$。然而，Muon并非沿着原始动量方向直接更新参数，而是沿着正交化动量的方向应用更新。具体而言，给定当前权重矩阵 $W_{t-1} \in \mathbb{R}^{m\times n}$ 和动量 $M_t \in \mathbb{R}^{m\times n}$，Muon的更新规则如下：

$$W_t = W_{t-1} - \eta\,\mathrm{Orth}(M_t)$$

其中 $UV^\top$ η $M_t = U\Sigma V^\top$ = 是学习率，而 $\mathrm{Orth}(M_t)$ 表示 M_t 的正交化，即使用 (薄) 奇异值分解 (SVD) 进行的 $\mathrm{Orth}(M_t)$。正交化过程通过将每个奇异值设为 1 来丢弃所有奇异值信息，同时保留由奇异向量诱导的方向结构。得到的正交化

matrix $UV^\top$ is commonly referred to as the *polar factor* (Higham, 2008; 2020) of M_t. In practice, however, Muon avoids the high cost of computing the full SVD by approximating the orthogonalization using efficient Newton–Schulz iterations (Jordan, 2024; Amsel et al., 2026).

NorMuon. NorMuon extends Muon by introducing row-wise adaptive scaling. In addition to the momentum buffer $M_t \in \mathbb{R}^{m\times n}$, it maintains a second-moment estimate $v_t \in \mathbb{R}^m$. Specifically, v_t tracks the row-wise mean square of the orthogonalized momentum. Rather than directly updating the parameters with the orthogonalized momentum, NorMuon first scales it by the inverse square root of v_t before applying the update. Formally, let $O_t = \mathrm{Orth}(M_t)$ and $\mathrm{mean}_{\mathrm{cols}}(\cdot)$ denote the row-wise mean, i.e., the mean across columns for each row. Given the decay rate $\beta_2 \in [0, 1)$ and the initialization $v_0 = \mathbf{0} \in \mathbb{R}^m$, the second-moment estimate v_t is computed via an exponential moving average (EMA)

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)\,\mathrm{mean}_{\mathrm{cols}}(O_t \odot O_t),$$

where $\odot$ denotes the Hadamard (element-wise) product. The final parameter update then uses O_t scaled by the inverse square root of v_t

$$W_t = W_{t-1} - \eta_t \frac{O_t}{\sqrt{v_t}+\delta}.$$

Here, the division is broadcast row-wise, η_t is the learning rate, and $\delta > 0$ is a small constant for numerical stability. For clarity, we omit the weight decay and learning rate scaling used in the original NorMuon, as neither affects the subsequent analysis.

3 On the Origin and Limitations of NorMuon's Adaptivity

Although NorMuon's row-wise adaptive scaling has demonstrated strong empirical performance, its underlying theoretical foundation remains largely unexplored.[1] In this section, we theoretically analyze the origin of NorMuon's adaptivity and uncover its inherent limitations. Specifically, we show that this adaptivity primarily arises from orthogonalization-induced geometry rather than optimization dynamics, and further demonstrate that such adaptivity can degenerate under certain conditions.

3.1 Exact Orthogonalization

We first consider the $m \le n$ regime, where the matrix is square ($m = n$) or wide ($m < n$). To facilitate the subsequent analysis, we express the row-wise mean of the element-wise square $O_t \odot O_t$ in terms of the Gram matrix of O_t

$$\mathrm{mean}_{\mathrm{cols}}(O_t \odot O_t) = \frac{1}{n}\,\mathrm{diag}(O_t O_t^\top),$$

where $\mathrm{diag}(\cdot)$ denotes the operator that extracts the diagonal entries of a square matrix into a vector. Recall that the thin SVD of M_t is given by $M_t = U\Sigma V^\top$. For $m \le n$, assuming that M_t has full row rank, we have $U \in \mathbb{R}^{m\times m}$, $\Sigma \in \mathbb{R}^{m\times m}$, and $V \in \mathbb{R}^{n\times m}$. By definition of the SVD, U is orthogonal ($UU^\top = U^\top U = I_m$) and V is column-orthonormal ($V^\top V = I_m$). Then, under exact orthogonalization, the identity $O_t = UV^\top$ holds exactly. Therefore, we obtain

$$O_t O_t^\top = (UV^\top)(UV^\top)^\top = U(V^\top V)U^\top = UU^\top = I_m,$$

i.e., O_t is row-orthonormal, and hence

$$\mathrm{mean}_{\mathrm{cols}}(O_t \odot O_t) = \frac{1}{n}\,\mathrm{diag}(O_t O_t^\top) = \frac{1}{n}\mathbf{1}_m,$$

where $\mathbf{1}_m$ denotes the all-ones vector in $\mathbb{R}^m$. Substituting this identity into the EMA update rule for v_t, unrolling the recursion, evaluating the resulting geometric series, and using $v_0 = \mathbf{0}$, we obtain

$$v_t = \beta_2^t v_0 + (1-\beta_2^t)\frac{1}{n}\mathbf{1}_m = (1-\beta_2^t)\frac{1}{n}\mathbf{1}_m,$$

which is uniform across rows, with only a global scalar varying over time. Furthermore, if the standard bias correction omitted in NorMuon were applied, it would cancel the time-dependent

[1] To the best of our knowledge, no theoretical convergence guarantee has yet been established for NorMuon.

矩阵 $UV^\top$ 通常被称为 M_t 的 极因子 (Higham, 2008; 2020)。然而，在实际应用中，Muon 通过使用高效的牛顿-舒尔茨迭代 (Jordan, 2024; Amsel等人, 2026) 来近似正交化，从而避免了计算完整 SVD 的高成本。

NorMuon. NorMuon 通过引入行自适应缩放扩展了 Muon。除了动量缓冲区 $M_t \in \mathbb{R}^{m\times n}$，它还维护一个二阶矩估计 $v_t \in \mathbb{R}^m$。具体来说， v_t跟踪正交化动量的行均方。NorMuon 不会直接用正交化动量更新参数，而是先将其乘以 v_t 的平方根的倒数，然后再应用更新。形式上，设 $O_t = \mathrm{Orth}(M_t)$ 和 $\mathrm{mean}_{\mathrm{cols}}(\cdot)$ 表示行均值，即每行的列均值。给定衰减率 $\beta_2 \in [0, 1)$ 和初始化 $v_0 = \mathbf{0} \in \mathbb{R}^m$， 二阶矩估计 v_t 通过指数移动平均（EMA）计算

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)\,\mathrm{mean}_{\mathrm{cols}}(O_t \odot O_t),$$

其中 $\odot$ 表示哈达玛（逐元素）积。最终的参数更新使用 O_t 乘以 v_t 的平方根的倒数

$$W_t = W_{t-1} - \eta_t \frac{O_t}{\sqrt{v_t}+\delta}.$$

在此，除法按行广播， η_t 是学习率， $\delta > 0$ 是用于数值稳定的较小常数。为清晰起见，我们省略了原始 NorMuon 中使用的权重衰减和学习率缩放，因为它们都不影响后续分析。

3 关于 NorMuon 自适应性的起源和局限性

尽管 NorMuon 的行自适应缩放已经展示了强大的经验性能，但其潜在的理论基础仍 largely 未被探索。[1] 在本节中，我们从理论上分析 NorMuon 自适应性的起源，并揭示其固有的局限性。具体来说，我们证明这种自适应性主要源于正交化引起的几何结构，而非优化动态，并进一步证明在特定条件下这种自适应性会退化。

3.1 精确正交化

我们首先考虑 $m \le n$ 情况，其中矩阵是方阵 ($m = n$) 或宽矩阵 ($m < n$)。为了便于后续分析，我们将元素平方的行均值 $O_t \odot O_t$ 表示为 O_t 的格拉姆矩阵。

$$\mathrm{mean}_{\mathrm{cols}}(O_t \odot O_t) = \frac{1}{n}\,\mathrm{diag}(O_t O_t^\top),$$

其中 $\mathrm{diag}(\cdot)$ 表示提取方阵对角线元素到向量的算子。回想一下， M_t 的薄奇异值分解 $M_t = U\Sigma V^\top$ 给出。对于 $m \le n$，假设 M_t 具有满行秩，我们有 $U \in \mathbb{R}^{m\times m}$、 $\Sigma \in \mathbb{R}^{m\times m}$ 和 $V \in \mathbb{R}^{n\times m}$。根据奇异值分解的定义， U 是正交 ($UU^\top = U^\top U = I_m$) 的，而 V 是列正交归一 ($V^\top V = I_m$) 的。然后，在精确正交化的情况下，恒等式 $O_t = UV^\top$ 精确成立。因此，我们得到

$$O_t O_t^\top = (UV^\top)(UV^\top)^\top = U(V^\top V)U^\top = UU^\top = I_m,$$

即， O_t 是行正交归一，因此

$$\mathrm{mean}_{\mathrm{cols}}(O_t \odot O_t) = \frac{1}{n}\,\mathrm{diag}(O_t O_t^\top) = \frac{1}{n}\mathbf{1}_m,$$

其中 $\mathbf{1}_m$ 表示 $\mathbb{R}^m$ 中的全1向量。将此恒等式代入 EMA更新规则 forv_t，展开递归，计算所得几何级数，并使用 $v_0 = \mathbf{0}$，我们得到

$$v_t = \beta_2^t v_0 + (1-\beta_2^t)\frac{1}{n}\mathbf{1}_m = (1-\beta_2^t)\frac{1}{n}\mathbf{1}_m,$$

这是在行内均匀的，只有全局标量随时间变化。此外，如果 NorMuon 中省略的标准偏差校正被应用，它会抵消随时间变化

[1] 据我们所知，NorMuon 尚未建立任何理论收敛保证。

factor, yielding exactly $\frac{1}{n}\mathbf{1}_m$, which is functionally equivalent to a fixed global scalar throughout training. The resulting scaling vector $\frac{1}{\sqrt{v_t}+\delta}$ therefore assigns the same scaling coefficient to each row. Hence, the originally intended neuron-wise adaptive scaling completely collapses to a trivial global scalar, thereby rendering the adaptive mechanism fundamentally ineffective.

3.2 Approximate Orthogonalization

The preceding analysis assumes exact orthogonalization. In practice, however, NorMuon employs approximate orthogonalization via a finite number of Newton–Schulz iterations (Li et al., 2026). We now analyze the approximate orthogonalization case. Assume that the approximately orthogonalized matrix O_t satisfies

$$O_t O_t^\top = I_m + E_t,$$

where $E_t \in \mathbb{R}^{m\times m}$ denotes the orthogonality residual induced by approximate orthogonalization. Then we have

$$\mathrm{mean}_{\mathrm{cols}}(O_t \odot O_t) = \frac{1}{n}\,\mathrm{diag}(O_t O_t^\top) = \frac{1}{n}\left(\mathbf{1}_m + \mathrm{diag}(E_t)\right).$$

Substituting this expression into the EMA update of v_t, unrolling the recursion, evaluating the resulting series, and using $v_0 = \mathbf{0}$, yields

$$v_t = (1-\beta_2^t)\frac{1}{n}\mathbf{1}_m + \frac{1-\beta_2}{n}\sum_{i=1}^{t}\beta_2^{t-i}\,\mathrm{diag}(E_i).$$

For convenience, define

$$R_t = \frac{1-\beta_2}{n}\sum_{i=1}^{t}\beta_2^{t-i}\,\mathrm{diag}(E_i).$$

The term $(1-\beta_2^t)\frac{1}{n}\mathbf{1}_m$ coincides exactly with the row-invariant scaling term derived in the exact orthogonalization case, whereas R_t captures the exponentially weighted accumulation of the orthogonality residuals and constitutes the only source of row-wise variation in v_t. To further characterize the magnitude of R_t, we assume that the orthogonality residual is uniformly bounded in spectral norm by some small constant $\epsilon > 0$, i.e., $\|E_t\|_2 \le \epsilon$ for all t. This assumption is consistent with practical Newton–Schulz implementations, where the orthogonalization error is controlled within a prescribed tolerance. The spectral norm bound directly implies a bound on the diagonal entries of E_t. Specifically, for each $j \in \{1,\dots,m\}$, by the Cauchy–Schwarz inequality, the definition of the induced operator norm, and the fact that $\|\mathbf{e}_j\|_2 = 1$, we have

$$|(E_t)_{jj}| = |\mathbf{e}_j^\top E_t \mathbf{e}_j| \le \|\mathbf{e}_j\|_2 \|E_t\mathbf{e}_j\|_2 \le \|E_t\|_2 \le \epsilon.$$

Consequently, we can bound R_t in the infinity norm as

$$\|R_t\|_\infty \le \frac{1-\beta_2}{n}\sum_{i=1}^{t}\beta_2^{t-i}\,\epsilon = (1-\beta_2^t)\frac{\epsilon}{n} \le \frac{\epsilon}{n}.$$

It immediately follows that the difference between any two entries of v_t is entirely determined by R_t. That is, for any $j,k \in \{1,\dots,m\}$, we have

$$|(v_t)_j - (v_t)_k| = |(R_t)_j - (R_t)_k| \le 2\|R_t\|_\infty \le \frac{2\epsilon}{n}.$$

This bounded variation in v_t directly translates into a bounded variation in the final adaptive scaling factors $\frac{1}{\sqrt{v_t}+\delta}$. Specifically, by the Mean Value Theorem, there exists some constant $C > 0$ such that, for any $j,k \in \{1,\dots,m\}$, we have

$$\left|\frac{1}{\sqrt{(v_t)_j}+\delta} - \frac{1}{\sqrt{(v_t)_k}+\delta}\right| \le \frac{C\epsilon}{n}.$$

In practice, Newton–Schulz-based orthogonalization rapidly reduces the orthogonality residual as iterations proceed, typically resulting in a very small ϵ. Consequently, the adaptive scaling factors

因子，从而得到精确的 $\frac{1}{n}\mathbf{1}_m$，这在功能上等效于在整个训练过程中保持固定的全局标量。因此，生成的缩放向量 $\frac{1}{\sqrt{v_t}+\delta}$ 为每一行分配相同的缩放系数。因此，原本打算的神经元自适应缩放完全崩溃为平凡的全局标量，从而使得自适应机制从根本上无效。

3.2 近似正交化

前面的分析假设了精确正交化。然而在实践中，NorMuon通过有限次数的牛顿-舒尔茨迭代采用近似正交化（Li等人，2026年）。我们现在分析近似正交化的情况。假设近似正交化的矩阵 O_t 满足

$$O_t O_t^\top = I_m + E_t,$$

其中 $E_t \in \mathbb{R}^{m\times m}$ 表示近似正交化引起的正交残差。那么我们有

$$\mathrm{mean}_{\mathrm{cols}}(O_t \odot O_t) = \frac{1}{n}\,\mathrm{diag}(O_t O_t^\top) = \frac{1}{n}\left(\mathbf{1}_m + \mathrm{diag}(E_t)\right).$$

将此表达式代入 v_t的EMA更新中，展开递归，计算结果序列，并使用 $v_0 = \mathbf{0}$，得到

$$v_t = (1-\beta_2^t)\frac{1}{n}\mathbf{1}_m + \frac{1-\beta_2}{n}\sum_{i=1}^{t}\beta_2^{t-i}\,\mathrm{diag}(E_i).$$

为方便起见，定义

$$R_t = \frac{1-\beta_2}{n}\sum_{i=1}^{t}\beta_2^{t-i}\,\mathrm{diag}(E_i).$$

该术语 $(1-\beta_2^t)\frac{1}{n}\mathbf{1}_m$ 与精确正交化情况中推导出的行不变缩放项完全一致，而 R_t 则捕获了正交残差的指数加权累积，并构成了 v_t中唯一的行变异性来源。为了进一步表征 R_t的幅度，我们假设正交残差在谱范数下被某个小的常数 $\epsilon > 0$统一界住，即 $\|E_t\|_2 \le \epsilon$ 对所有 t 成立。这一假设与实际牛顿-舒尔茨实现一致，其中正交化误差被控制在规定的容差范围内。谱范数界直接意味着E_t的对角线元素有界。具体而言，对于每个 $j \in \{1,\dots,m\}$，根据柯西-施瓦茨不等式、诱导算子范数的定义以及 $\|\mathbf{e}_j\|_2 = 1$这一事实，我们有

$$|(E_t)_{jj}| = |\mathbf{e}_j^\top E_t \mathbf{e}_j| \le \|\mathbf{e}_j\|_2 \|E_t\mathbf{e}_j\|_2 \le \|E_t\|_2 \le \epsilon.$$

因此，我们可以将 R_t 在无穷范数下界定为

$$\|R_t\|_\infty \le \frac{1-\beta_2}{n}\sum_{i=1}^{t}\beta_2^{t-i}\,\epsilon = (1-\beta_2^t)\frac{\epsilon}{n} \le \frac{\epsilon}{n}.$$

由此立即可知， v_t 中任意两个条目的差值完全由R_t决定。也就是说，对于任意 $j,k \in \{1,\dots,m\}$，我们有

$$|(v_t)_j - (v_t)_k| = |(R_t)_j - (R_t)_k| \le 2\|R_t\|_\infty \le \frac{2\epsilon}{n}.$$

在 v_t 中的有界变差直接转化为最终自适应缩放因子 $\frac{1}{\sqrt{v_t}+\delta}$的有界变差。具体来说，根据中值定理，存在某个常数 $C > 0$，使得对于任意 $j,k \in \{1,\dots,m\}$，我们有

$$\left|\frac{1}{\sqrt{(v_t)_j}+\delta} - \frac{1}{\sqrt{(v_t)_k}+\delta}\right| \le \frac{C\epsilon}{n}.$$

实际上，基于牛顿-舒尔茨的正交化在迭代过程中能快速减小正交残差，通常导致 ϵ非常小。因此，自适应缩放因子

exhibit only negligible variation across rows, rendering the row-wise adaptivity practically ineffective.

In short, the above analysis reveals that, in the $m \leq n$ regime, the apparent row-wise adaptivity in NorMuon is not driven by the underlying gradient statistics or curvature information that adaptive optimizers are typically designed to capture. Instead, the remaining row-wise variation in the adaptive scaling factors is solely induced by the orthogonalization residual. In other words, the observed adaptivity of NorMuon is merely a byproduct of inexact orthogonalization. As $\epsilon \to 0$, this residual-induced adaptivity vanishes, and the behavior of approximate orthogonalization converges to that of exact orthogonalization, recovering uniform, non-adaptive row-wise scaling.

Remark. A counterintuitive implication of the preceding analysis, which we term the ***Orthogonalization–Adaptivity Paradox***, is that the adaptivity of NorMuon is inversely coupled with the quality of orthogonalization. The more accurate the orthogonalization, the weaker the adaptivity becomes; in the limit of exact orthogonalization, the adaptivity disappears entirely. Conversely, less accurate orthogonalization permits greater variation in the row-wise scaling factors and thus allows for stronger adaptivity. This suggests that NorMuon's adaptive mechanism essentially serves to compensate for imperfections in orthogonalization, rather than providing genuine optimization-driven adaptivity. This stands in stark contrast to the design philosophy of Muon, where accurate orthogonalization is the cornerstone of the optimizer. Consequently, NorMuon's adaptive mechanism is intrinsically incompatible with ongoing research efforts to improve Muon through increasingly accurate orthogonalization (Grishina et al., 2025; He et al., 2025; Amsel et al., 2026).

Note that the above analysis relies on the full row rank assumption. In fact, this already covers the generic case encountered in practice. Rank deficiency is an undesirable edge case that can hinder effective training and impair the model's representational capacity, and is thus generally avoided in practice. Since it rarely occurs, we do not treat it separately here or in the subsequent analysis. We do emphasize, however, that even under rank deficiency, the row-wise scaling of NorMuon remains entirely determined by the orthogonalization procedure rather than by the optimization dynamics. A more detailed discussion of this case will be provided in an extended version of this work.

3.3 THE $m > n$ REGIME

In the $m > n$ regime, where the matrix is tall, the situation is fundamentally different. The row-orthonormality property $O_t O_t^\top = I_m$ no longer holds due to the dimensional constraint $m > n$. Instead, under exact orthogonalization, the column-orthonormality property $O_t^\top O_t = I_n$ holds. This implies that every column of O_t has unit norm, i.e., $\|(O_t)_{:,j}\|_2^2 = 1$ for each $j \in \{1, \ldots, n\}$. In contrast, since $O_t O_t^\top \neq I_m$, the squared norm of each row of O_t is no longer constrained to 1. Nevertheless, the aggregate row energy is constrained to be fixed. Specifically, we have

$$\sum_{i=1}^{m} \|(O_t)_{i,:}\|_2^2 = \sum_{i=1}^{m}\sum_{j=1}^{n} (O_t)_{ij}^2 = \sum_{j=1}^{n}\sum_{i=1}^{m} (O_t)_{ij}^2 = \sum_{j=1}^{n} \|(O_t)_{:,j}\|_2^2 = n,$$

where the last equality follows from the column-orthonormality of O_t. Since there is no row orthonormality constraint, this fixed aggregate energy is no longer forced to be uniformly distributed across the m rows. Consequently, the squared norm of each row can deviate substantially from the uniform allocation n/m, inducing row-wise variation in $\mathrm{mean}_{\mathrm{cols}}(O_t \odot O_t)$ and hence in the resulting adaptive scaling factors. This indicates that, in the $m > n$ regime, NorMuon's adaptivity is not driven by conventional optimization dynamics underlying most existing adaptive optimizers, but instead stems from the intrinsic uneven row-energy distribution induced by tall-matrix orthogonalization.

The above analysis focuses on exact orthogonalization. For approximate orthogonalization, the analysis largely follows that of the $m \leq n$ regime, where deviations from exact orthogonality further introduce residual-induced variation in the adaptive scaling factors. This residual effect also diminishes as the orthogonalization accuracy improves, following the same Orthogonalization–Adaptivity Paradox.

Summary. Our analysis in the three preceding subsections reveals that NorMuon's adaptivity is not optimization-driven in the conventional sense. Instead, it primarily serves as a compensation

在各行之间仅表现出微不足道的差异，使得按行自适应实际上无效。

简而言之，上述分析表明，在 $m \leq n$ 机制下，NorMuon中的表观行缩放适应性并非由自适应优化器通常设计用来捕获的底层梯度统计或曲率信息驱动。相反，自适应缩放因子中剩余的行变化完全由正交化残差引起。换句话说，NorMuon的观察到的适应性仅仅是精确正交化的副产品。正如 $\epsilon \to 0$，这种残差诱导的适应性消失，近似正交化的行为收敛到精确正交化的行为，恢复均匀、非自适应的行缩放。

注意。上述分析的一个反直觉推论，我们称之为**正交化-适应性悖论**，是NorMuon的适应性与其正交化质量成反比。正交化越准确，适应性越弱；在精确正交化的极限下，适应性完全消失。相反，不太准确的正交化允许行缩放因子有更大的变化，从而允许更强的适应性。这表明NorMuon的自适应机制本质上是为了弥补正交化中的不完美，而不是提供真正的优化驱动适应性。这与Muon的设计理念形成鲜明对比，在Muon中，精确正交化是优化器的基石。因此，NorMuon的自适应机制本质上与通过日益精确的正交化来改进Muon的持续研究工作不相容（Grishina等人，2025年；He等人，2025年；Amsel等人，2026年）。

请注意，上述分析依赖于满行秩假设。事实上，这已经涵盖了实践中遇到的通用情况。秩亏是一种不受欢迎的边缘情况，它可能会阻碍有效训练并损害模型的表征能力，因此在实践中通常被避免。由于它很少发生，我们在这里或后续分析中都不单独处理它。然而，我们确实强调，即使在秩亏的情况下，NorMuon 的行缩放仍然完全由正交化过程决定，而不是由优化动力学决定。在本工作的扩展版本中，将提供对此情况的更详细讨论。

3.3 THE $m > n$ REGIME

在 $m > n$ 模式下，当矩阵较宽时，情况根本不同。由于维数限制 $m > n$，行正交规范性 $O_t O_t^\top = I_m$ 不再成立。相反，在精确正交化下，列正交规范性 $O_t^\top O_t = I_n$ 成立。这意味着 O_t 的每一列都具有单位范数，即 $\|(O_t)_{:,j}\|_2^2 = 1$ 对每个 $j \in \{1, \ldots, n\}$。相比之下，由于 $O_t O_t^\top \neq I_m$，O_t 的每一行的平方范数不再被约束为1。尽管如此，总行能量被约束为固定值。具体来说，我们有

$$\sum_{i=1}^{m} \|(O_t)_{i,:}\|_2^2 = \sum_{i=1}^{m}\sum_{j=1}^{n} (O_t)_{ij}^2 = \sum_{j=1}^{n}\sum_{i=1}^{m} (O_t)_{ij}^2 = \sum_{j=1}^{n} \|(O_t)_{:,j}\|_2^2 = n,$$

其中最后一个等式源于 O_t 的列正交规范性。由于没有行正交规范性约束，这个固定的总能量不再被强制均匀分布在 m 行中。因此，每一行的平方范数可以显著偏离均匀分配 n/m，导致行内均值 $_{\mathrm{cols}}(O_t \odot O_t)$ 的变化，进而影响最终的自适应缩放因子。这表明，在 $m > n$ 模式下，NorMuon 的自适应性并非由大多数现有自适应优化器背后的传统优化动态驱动，而是源于高维矩阵正交化所诱导的内在不均匀行能量分布。

上述分析侧重于精确正交化。对于近似正交化，分析大体上遵循 $m \leq n$ 模式的框架，其中偏离精确正交性会进一步引入残差引起的自适应缩放因子变化。这种残差效应也会随着正交化精度的提高而减弱，遵循相同的正交化-适应性悖论。

摘要。我们前三个小节的分析表明，NorMuon的自适应性并非传统意义上的优化驱动。相反，它主要作为补偿

Algorithm 1 Decoupled Geometry-Aligned Adaptive Muon (**DGA-Muon**)

Input: Step size $\eta > 0$, decay rates $\beta_1, \beta_2 \in [0, 1)$, stabilizer $\delta > 0$, clip ratios $c \in (0, 1], d \geq 1$
Initialize: $W_1 \in \mathbb{R}^{m\times n}$, $M_0 = \mathbf{0} \in \mathbb{R}^{m\times n}$, $v_0 = \mathbf{0} \in \mathbb{R}^{\min(m,n)}$
for $t = 1, \ldots, T$ **do**
  Compute stochastic gradient G_t
  Update first moment estimate $M_t = \beta_1 M_{t-1} + (1 - \beta_1)G_t$
  Update second moment estimate from raw gradients

$$v_t = \begin{cases} \beta_2 v_{t-1} + (1 - \beta_2)\,\mathrm{diag}(G_t G_t^\top), & m \leq n, \\ \beta_2 v_{t-1} + (1 - \beta_2)\,\mathrm{diag}(G_t^\top G_t), & m > n. \end{cases}$$

  Apply bias correction $\hat{v}_t = v_t/(1 - \beta_2^t)$
  Compute adaptive scaling factors $s_t = 1/(\sqrt{\hat{v}_t} + \delta)$
  Compute median of scaling factors $s_{\text{med}} = \mathrm{median}(s_t)$
  Clip scaling factors $s_t \leftarrow \mathrm{clip}(s_t,\, c \cdot s_{\text{med}},\, d \cdot s_{\text{med}})$
  Orthogonalize first moment $O_t = \mathrm{Orth}(M_t)$
  Apply geometry-aligned adaptive scaling

$$\widetilde{O}_t = \begin{cases} \mathrm{Diag}(s_t)\, O_t, & m \leq n, \\ O_t\, \mathrm{Diag}(s_t), & m > n. \end{cases}$$

  Update parameters $W_{t+1} = W_t - \eta \widetilde{O}_t$
end for

mechanism for the non-uniformity induced by orthogonalization, including residual errors and uneven row-energy distribution. Although this compensation can provide empirical benefits, the resulting adaptivity is largely an orthogonalization-induced *byproduct* rather than a principled adaptive mechanism. Furthermore, approximate orthogonalization gives rise to the Orthogonalization–Adaptivity Paradox, revealing an inherent limitation of this form of adaptation. These findings motivate the design of a more principled adaptive mechanism tailored specifically to Muon.

4 Decoupled Geometry-Aligned Adaptive Scaling

In this section, we present a novel adaptive scaling mechanism specifically designed for Muon. We first outline the core design principles and then formally justify the orthogonality-preserving property of the resulting scaling strategy. We further discuss additional design considerations for constructing the adaptive scaling factors. The complete optimization procedure is summarized in Algorithm 1[2].

4.1 Design Principles: Decoupled Adaptivity and Orthogonality Preservation

We begin by examining the root cause of the limitations of NorMuon's adaptivity identified in the preceding section. Specifically, our analysis reveals that these limitations stem from the inherent coupling between orthogonalization and adaptive scaling. Because the scaling factors are computed from the orthogonalized momentum, the resulting adaptivity is inherently dictated by orthogonalization-induced geometry rather than direct gradient information. This analysis naturally motivates a core design principle: the computation of adaptive scaling factors should be decoupled from the orthogonalization procedure. Guided by this principle, we propose a decoupled adaptive mechanism that computes the adaptive scaling factors directly from the raw gradients, rather than from the orthogonalized momentum, while applying these factors to the orthogonalized update, i.e., the polar factor. In this way, the adaptive scaling is no longer dictated by orthogonalization-induced geometry, but instead reflects genuine optimization-relevant information—a defining characteristic of standard adaptive optimizers.

[2]We omit the weight decay term here for generality, as it is independent of the core algorithm and various variants can be seamlessly incorporated (e.g., standard or cautious weight decay (Chen et al., 2026)).

算法1解耦几何对齐自适应Muon (**DGA-Muon**)

1: **输入：** 步长 $\eta > 0$，衰减率 β_1，$\beta_2 \in [0, 1)$，稳定器 $\delta > 0$，裁剪比例 $c \in (0,\ 1]$，$d \geq 1$
2: **初始化：** $W_1 \in \mathbb{R}^{m\times n}$, $M_0 = \mathbf{0} \in \mathbb{R}^{m\times n}$, $v_0 = \mathbf{0} \in \mathbb{R}^{\min(m,n)}$3: **for** $t = 1,\ \ldots,\ T$**do**4: 计算随机梯度 G_t5: 更新第一矩估计 $M_t = \beta_1 M_{t-1} +(1 - \beta_1)G_t$6: 从原始梯度更新第二矩估计

$$v_t = \begin{cases} \beta_2 v_{t-1} + (1 - \beta_2)\,\mathrm{diag}(G_t G_t^\top), & m \leq n, \\ \beta_2 v_{t-1} + (1 - \beta_2)\,\mathrm{diag}(G_t^\top G_t), & m > n. \end{cases}$$

7: 应用偏差校正 $\hat{v}_t = v_t/(1 - \beta_2^t)$
8: 计算自适应缩放因子 $s_t = 1/(\sqrt{\hat{v}_t} + \delta)$
9: 计算缩放因子的中位数 $s_{\text{med}} = \mathrm{median}(s_t)$
10: 裁剪缩放因子 $s_t \leftarrow \mathrm{clip}(s_t,\, c \cdot s_{\text{med}},\, d \cdot s_{\text{med}})$
11: 对第一个矩进行正交化 $O_t = \mathrm{Orth}(M_t)$
12: 应用几何对齐的自适应缩放

$$\widetilde{O}_t = \begin{cases} \mathrm{Diag}(s_t)\, O_t, & m \leq n, \\ O_t\, \mathrm{Diag}(s_t), & m > n. \end{cases}$$

13: 更新参数 $W_{t+1} = W_t - \eta O_t$14: **结束循环**

正交化引起的非均匀性补偿机制，包括残差误差和不均匀的行能量分布。尽管这种补偿可以提供经验性收益，但结果的自适应性主要是由正交化产生的<样式 id='1'>副产品</样式>，而不是一种原则性的自适应机制。此外，近似正交化导致了正交化-自适应性悖论，揭示了这种适应形式固有的局限性。这些发现促使我们设计了一种更原则性的自适应机制，专门针对Muon进行优化。

4 解耦几何对齐自适应缩放

在本节中，我们提出了一种专门为Muon设计的自适应缩放机制。我们首先概述了核心设计原则，然后正式证明了所得缩放策略的正交性保持特性。我们进一步讨论了构建自适应缩放因子的其他设计考虑。完整的优化过程总结在算法 1[2]中。

4.1 设计原则：解耦自适应性与正交性保持

我们首先检查 NorMuon 自适应性的局限性，这些局限性在上一节中已被识别。具体而言，我们的分析表明，这些局限性源于正交化和自适应缩放之间的固有耦合。由于缩放因子是从正交化动量计算的，因此产生的自适应性本质上是由正交化引起的几何形状决定的，而不是由直接梯度信息决定的。这项分析自然地引出了一个核心设计原则：自适应缩放因子的计算应与正交化过程解耦。根据这一原则，我们提出了一种解耦的自适应机制，该机制直接从原始梯度计算自适应缩放因子，而不是从正交化动量计算，同时将这些因子应用于正交化更新，即极因子。通过这种方式，自适应缩放不再受正交化引起的几何形状决定，而是反映真实的优化相关信息——这是标准自适应优化器的定义特征。

[2]我们在此为通用性省略了权重衰减项，因为它独立于核心算法，且各种变体可以无缝集成（例如，标准或谨慎的权重衰减（Chen等人，2026年））。

While decoupling the computation of adaptive scaling factors resolves the fundamental coupling between adaptivity and orthogonalization, a closer examination of the geometric structure of the orthogonalized update reveals another design deficiency in NorMuon. Specifically, in the $m > n$ regime, the polar factor has orthonormal columns, which define the orthogonal directions of the update, whereas NorMuon applies adaptive scaling along the row dimension. This dimensional mismatch inevitably distorts these orthogonal directions and destroys the orthogonality that the polar factor is designed to preserve. This observation motivates a second design principle: adaptive scaling should respect the intrinsic geometry of the orthogonalized update. Intuitively, since orthogonalization is designed to extract a set of orthogonal update directions, any subsequent adaptive scaling should preserve this directional structure rather than distort it. Following this principle, we introduce a geometry-aligned scaling mechanism that applies row-wise scaling for $m < n$ and column-wise scaling for $m > n$, while either choice can be used when $m = n$, thereby preserving the orthogonality of the update in all regimes.

4.2 Formal Justification of the Orthogonality-Preserving Property

We now provide a formal justification of the orthogonality-preserving property for the proposed geometry-aligned scaling mechanism. We first consider the $m > n$ regime under exact orthogonalization, where the polar factor $O_t \in \mathbb{R}^{m\times n}$ satisfies $O_t^\top O_t = I_n$. In this setting, the columns of O_t form a set of mutually orthogonal directions. Let $v_t \in \mathbb{R}^n$ denote the column-wise second-moment estimate, where each entry tracks the EMA of the squared ℓ_2-norm of the corresponding column of the raw gradients, i.e.,

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)\,\mathrm{diag}\big(G_t^\top G_t\big)\,.$$

Note that the Gram matrix here is introduced only for notational convenience; in practice, only its diagonal entries are needed, so the full Gram matrix is never explicitly formed. The associated diagonal scaling matrix is constructed as $H_t = \mathrm{Diag}(1/(\sqrt{v_t}+\delta)) \in \mathbb{R}^{n\times n}$, where $\mathrm{Diag}(\cdot)$ denotes the operator that constructs a diagonal matrix from a vector. Applying column-wise adaptive scaling to O_t is equivalent to right-multiplying O_t by H_t, yielding the scaled update $\widetilde{O}_t = O_t H_t$. The Gram matrix of the scaled update satisfies

$$\widetilde{O}_t^\top \widetilde{O}_t = H_t^\top O_t^\top O_t H_t = H_t^\top I_n H_t = H_t^2,$$

which remains diagonal. This implies that the columns of $\widetilde{O}_t$ remain mutually orthogonal, i.e.,

$$\langle \widetilde{O}_{:,i},\, \widetilde{O}_{:,j}\rangle = 0,\ i \neq j,$$

while their squared norms satisfy

$$\|\widetilde{O}_{:,i}\|_2^2 = (H_t^2)_{ii},$$

indicating that the magnitude of each orthogonal direction is independently scaled by its corresponding adaptive scaling factor. Thus, column-wise adaptive scaling transforms the polar factor with unit column norms into an update with orthogonal columns but heterogeneous column norms. It preserves the orthogonal directions extracted by orthogonalization while assigning an adaptive step size to each direction. In contrast, applying row-wise adaptive scaling to O_t is equivalent to left-multiplying O_t by the corresponding diagonal scaling matrix $D_t \in \mathbb{R}^{m\times m}$, constructed analogously from the row-wise second-moment estimate, yielding the scaled update $\widetilde{O}_t = D_t O_t$. The resulting Gram matrix becomes

$$\widetilde{O}_t^\top \widetilde{O}_t = O_t^\top D_t^\top D_t O_t = O_t^\top D_t^2 O_t,$$

which is generally non-diagonal. This indicates that row-wise adaptive scaling reintroduces coupling between the orthogonal column directions extracted by orthogonalization, and hence the orthogonal structure established by orthogonalization is no longer preserved in the $m > n$ regime.

For the $m < n$ regime, the analysis proceeds analogously, with the roles of rows and columns reversed. In this case, we construct the adaptive scaling factors using the row-wise second-moment estimate $v_t \in \mathbb{R}^m$, where each entry tracks the EMA of the squared ℓ_2-norm of the corresponding row of the raw gradients, i.e.,

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)\,\mathrm{diag}(G_t G_t^\top).$$

The polar factor O_t satisfies $O_t O_t^\top = I_m$, implying that the rows of O_t form a set of mutually orthogonal directions. Applying row-wise adaptive scaling to O_t is equivalent to left-multiplying

在解耦自适应缩放因子的计算解决了自适应性与正交化之间的基本耦合问题后，对正交化更新几何结构的更仔细检查揭示了 NorMuon 中的另一个设计缺陷。具体来说，在 $m > n$ 机制下，极因子具有正交归一列，这些列定义了更新的正交方向，而 NorMuon 沿行维度应用自适应缩放。这种维度不匹配不可避免地扭曲了这些正交方向，并破坏了极因子旨在保留的正交性。这一观察结果促使了第二个设计原则：自适应缩放应尊重正交化更新的固有几何结构。直观地讲，由于正交化旨在提取一组正交更新方向，任何后续的自适应缩放都应保留这种方向结构而不是扭曲它。遵循这一原则，我们引入了一种几何对齐缩放机制，该机制对 $m < n$ 应用行缩放，对 $m > n$ 应用列缩放，而在 $m = n$ 时两者选择均可，从而在所有机制下都保持了更新的正交性。

4.2 保持正交性形式的正式证明

我们现在为所提出的几何对齐缩放机制提供保持正交性的正式证明。我们首先考虑精确正交化的 $m > n$ 情况，其中极因子 $O_t \in \mathbb{R}^{m\times n}$ 满足 $O_t^\top O_t = I_n$。在此设置中， O_t的列形成一组相互正交的方向。令 $v_t \in \mathbb{R}^n$ 表示按列的二阶矩估计，其中每个条目跟踪原始梯度相应列的平方 ℓ_2-范数的EMA，即，

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)\,\mathrm{diag}\big(G_t^\top G_t\big)\,.$$

请注意，这里的格拉姆矩阵仅用于符号方便；在实践中，仅需要其对角线元素，因此完整的格拉姆矩阵从未显式形成。相关的对角线缩放矩阵按 $H_t = \mathrm{Diag}(1/(\sqrt{v_t}+\delta)) \in \mathbb{R}^{n\times n}$构造，其中Diag(·)表示从向量构造对角矩阵的算子。将列自适应缩放$\widetilde{\ }$应用于 O_t 相当于用 O_t 右乘 H_t，得到缩放更新 $\widetilde{O}_t = O_t H_t$。缩放更新的格拉姆矩阵满足

$$\widetilde{O}_t^\top \widetilde{O}_t = H_t^\top O_t^\top O_t H_t = H_t^\top I_n H_t = H_t^2,$$

$\widetilde{\ }$保持对角线。这意味着 O_t 的列保持相互正交，即，

$$\langle \widetilde{O}_{:,i},\, \widetilde{O}_{:,j}\rangle = 0,\ i \neq j,$$

而它们的平方范数满足

$$\|\widetilde{O}_{:,i}\|_2^2 = (H_t^2)_{ii},$$

表明每个正交方向的幅度由其对应的自适应缩放因子独立缩放。因此，列自适应缩放将具有单位列范数的极因子转换为具有正交列但异构列范数的更新。它保留了正交化提取的正交方向，同时为每个方向分配自适应步长。相比之下，将行自适应缩放应用于 O_t 相当于用相应的对角线缩放矩阵 $D_t \in \mathbb{R}^{m\times m}$（类似地由行二阶矩估计构建）左乘 O_t，得到缩放后的更新 $\widetilde{O}_t = D_t O_t$。得到的格拉姆矩阵变为

$$\widetilde{O}_t^\top \widetilde{O}_t = O_t^\top D_t^\top D_t O_t = O_t^\top D_t^2 O_t,$$

这通常是 非 对角线 的。这表明行自适应缩放重新引入了正交化提取的正交列方向之间的耦合，因此正交化建立的正交结构在 $m > n$ 情况下不再保持。

对于 $m < n$ 模式，分析过程类似，只是行和列的角色互换了。在这种情况下，我们使用按行计算的二阶矩估计 $v_t \in \mathbb{R}^m$ 构建自适应缩放因子，其中每个元素跟踪原始梯度对应行的平方 ℓ_2-范数的 EMA，即，

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)\,\mathrm{diag}(G_t G_t^\top).$$

极因子 O_t 满足 $O_t O_t^\top = I_m$，这意味着 O_t 的行构成一组相互正交的方向。将行自适应缩放应用于 O_t 相当于左乘

O_t by the corresponding diagonal scaling matrix, thereby preserving the row-wise orthogonal structure. In contrast, applying column-wise adaptive scaling, equivalently right-multiplying O_t by the corresponding diagonal scaling matrix, generally distorts the row-wise orthogonal structure. For the $m = n$ regime, the polar factor O_t is a square orthogonal matrix, satisfying both $O_t O_t^\top = I_m$ and $O_t^\top O_t = I_m$. Hence, both the rows and columns of O_t form sets of mutually orthogonal directions, and either row-wise or column-wise adaptive scaling preserves the corresponding orthogonal directions, adjusting only their magnitudes.

Remark: Broader Insights for Matrix-Aware Optimizer Design. The design principles and geometric justifications presented above provide deeper insights into the general paradigm of matrix-aware optimization.

1. **A Generalized Design Philosophy.** Our decoupled, direction-aware adaptive mechanism transcends being a mere enhancement for Muon; it constitutes a more general design philosophy for matrix-aware optimizers. Notably, the directions utilized need not be restricted to the orthogonal directions used by Muon, but can naturally extend to broader classes of directions that characterize matrices.
2. **Intrinsic Necessity of Row-Wise/Column-Wise Adaptivity.** This perspective formally establishes why row-wise or column-wise adaptivity is fundamentally necessary: it provides a natural mechanism for adapting to the one-sided orthogonal directions of matrices. Rather than viewing row-wise/column-wise scaling as a structural approximation to Adam-like element-wise scaling (as in Adam-mini (Zhang et al., 2025b) and other related work), our approach grounds it in the intrinsic geometry of matrices (Zhang et al., 2026b). Furthermore, although methods such as NorMuon also employ row-wise/column-wise adaptation, our motivation and design arise from a distinct geometric alignment principle rather than heuristic normalization.

4.3 Additional Design Considerations

Beyond the two aforementioned design principles, we further incorporate several additional considerations when constructing the adaptive scaling factors to better regulate their dynamics during optimization. First, the second-moment estimate is constructed from the sum rather than the mean of the squared entries. While this is already implicit in the construction of v_t above, we make this design choice more explicit here. For a row or column gradient vector, the corresponding second-moment term is given by its squared ℓ_2-norm, without normalization by the vector dimension. This reflects our design philosophy of treating each row (or column) gradient vector as a single adaptive unit, with its scaling determined by the total accumulated gradient energy of the vector. Consequently, units with larger accumulated gradient energy receive smaller scaling factors, while units with lower accumulated energy are assigned larger ones. This contrasts with NorMuon, which normalizes the row-wise gradient energy by the row dimension, making its adaptive scaling depend on the average per-coordinate energy rather than the total energy of each row.

Since the second-moment estimate v_t is computed directly from raw gradients and is decoupled from orthogonalization, it suffers from the standard initialization bias of EMA estimators. In particular, during the early optimization steps, v_t tends to underestimate the underlying second moment due to the zero initialization of the accumulator. We therefore apply the standard bias-correction technique to obtain a bias-corrected estimate $\hat{v}_t = \frac{v_t}{1-\beta_2^t}$, which is then used to compute the adaptive scaling factor $s_t = \frac{1}{\sqrt{\hat{v}_t}+\delta}$.[3] Here v_t, $\hat{v}_t$, and s_t are generic notations that can represent vectors in either $\mathbb{R}^m$ or $\mathbb{R}^n$. Note that the first moment M_t requires no bias correction, owing to the scale-invariance of the subsequent orthogonalization operation (Pethick et al., 2025).

While introducing adaptive scaling into Muon provides more fine-grained control over different adaptive units, it may also lead to excessive anisotropic scaling, resulting in overly imbalanced updates across different orthogonal directions. Such excessive anisotropy necessitates the use of smaller step sizes to maintain stable optimization. To preserve Muon's favorable property of supporting large step sizes, we introduce a clipping operation that bounds the degree of adaptivity. Specifically, we apply the following element-wise clipping operation

$$\mathrm{clip}(s_t, s_{t,\min}, s_{t,\max}) = \min(\max(s_t, s_{t,\min}),\ s_{t,\max}).$$

[3]Bias correction can be replaced with a gradient normalization technique, which we will include in revision.

O_t 由相应的对角线缩放矩阵，从而保留行正交结构。相比之下，应用列自适应缩放，相当于右乘 O_t 由相应的对角线缩放矩阵，通常会扭曲行正交结构。对于$m = n$ 机制，极因子 O_t 是一个正交方阵，满足 $O_t O_t^\top = I_m$ 和$O_t^\top O_t = I_m$。因此， O_t 的行和列都构成一组相互正交的方向，无论是行自适应缩放还是列自适应缩放，都会保留相应的正交方向，仅调整其大小。

备注：为矩阵感知优化器设计提供更广泛的见解。 上述提出的设计原则和几何合理性为矩阵感知优化的通用范式提供了更深入的见解。

1. **广义设计理念**。 我们的解耦、方向感知的自适应机制超越了仅仅是对Muon的增强；它构成了矩阵感知优化器更一般的 设计理念。值得注意的是，所使用的方向不必局限于Muon使用的正交方向，但可以自然地扩展到表征矩阵更广泛的方向类。

2. **行/列自适应的内在必要性**。 这种视角正式确立了为什么行或列的自适应是根本必要的：它为适应矩阵的单侧正交方向提供了自然机制。我们的方法不是将行/列缩放视为Adam类元素级缩放的结构近似（如在Adam-mini (Zhang等人, 2025b) 和其他相关工作 中），而是将其建立在矩阵的固有几何结构（Zhang等人, 2026b）之上。此外，尽管NorMuon等方 法也采用行/列自适应，但我们的动机和设计源于不同的几何对齐原则，而非启发式归一化。

4.3 附加设计考虑

在上述两种设计原则之外，我们在构建自适应缩放因子时进一步考虑了若干附加因素，以更好地在优化过程中调节其动态特性。首先，二阶矩估计是通过平方项的总和而非均值构建的。虽然这一点在上述 v_t 的构建中已经隐含，但我们在此处更明确地做出这一设计选择。对于行或列梯度向量，相应的二阶矩项由其平方 ℓ_2-范数给出，且不进行向量维度的归一化。这反映了我们设计理念中，将每个行（或列）梯度向量视为一个单一自适应单元，其缩放由向量的总累积梯度能量决定。因此，累积梯度能量较大的单元会获得较小的缩放因子，而累积能量较低的单元则被分配较大的缩放因子。这与 NorMuon 形成对比，后者通过行维度对行梯度能量进行归一化，使其自适应缩放依赖于每个坐标的平均能量而非每行的总能量。

由于二阶矩估计 v_t 直接从原始梯度计算且与正交化解耦，因此它受到EMA估计器的标准初始化偏差的影响。特别是，在早期优化步骤中， v_t 由于累加器的零初始化而倾向于低估潜在的二阶矩。因此，我们应用标准的偏差校正技术以获得偏差校正估计$\hat{v}_t = \frac{v_t}{1-\beta_2^t}$， 然后使用该估计来计算自适应缩放 —— 因子 $s_t = \frac{1}{\sqrt{\hat{v}_t}+\delta}$.[3] 这里 v_t、$\hat{v}_t$和 s_t 是通用符号，可以表示 $\mathbb{R}^m$或 $\mathbb{R}^n$中的向量。请注意，一阶矩 M_t 无需偏差校正，因为后续的正交化操作具有尺度不变性（Pethick等人，2025年）。

在 Muon 中引入自适应缩放虽然能提供更细粒度的控制，但可能导致各向异性缩放过度，从而在不同正交方向上产生过于不平衡的更新。这种过度的各向异性需要使用更小的步长来维持稳定的优化。为了保留 Muon 支持大步长的优点，我们引入了一种裁剪操作来限制自适应程度。具体来说，我们应用以下逐元素裁剪操作

$$\mathrm{clip}(s_t, s_{t,\min}, s_{t,\max}) = \min(\max(s_t, s_{t,\min}),\ s_{t,\max}).$$

[3]偏差校正可以用梯度归一化技术替代，我们将在修订版中包含该技术。

We determine the clipping bounds $s_{t,\min}$ and $s_{t,\max}$ adaptively based on the median of the scaling factors, denoted by $s_{\text{med}} = \text{median}(s_t)$. Specifically, we set $s_{t,\min} = c \cdot s_{\text{med}}$ and $s_{t,\max} = d \cdot s_{\text{med}}$, where $c \in (0, 1]$ and $d \geq 1$ control the allowable dynamic range. One can also use a single hyperparameter by setting $d = 1/c$. The clipping operation effectively bounds the condition number of the induced diagonal scaling matrix and prevents disproportionately large or small updates along specific orthogonal directions. When $s_{t,\min} = s_{t,\max}$, the clipping removes all relative adaptivity, reducing the update to a uniformly scaled Muon update.

5 Theoretical Analysis

In this section, we establish the convergence guarantee of DGA-Muon under exact orthogonalization. We first introduce the assumptions required for the analysis.

Assumptions. We make the following assumptions throughout the analysis.

(A1) (L-smoothness) $\|\nabla f(X) - \nabla f(Y)\|_F \leq L\|X - Y\|_F$, for all X, Y.

(A2) (Unbiased stochastic gradients with bounded variance) The stochastic gradients satisfy

$$\mathbb{E}[G_t \mid \mathcal{F}_{t-1}] = \nabla f(W_t), \quad \mathbb{E}\|G_t - \nabla f(W_t)\|_F^2 \leq \sigma^2.$$

(A3) (Bounded stochastic gradients) There exists a constant $G_{\max} > 0$ such that

$$\|G_t\|_F \leq G_{\max} \text{ for all } t.$$

We now proceed with the convergence analysis. By the L-smoothness of f and the update rule $W_{t+1} = W_t - \eta \widetilde{O}_t$, we have

$$f(W_{t+1}) \leq f(W_t) + \langle \nabla f(W_t), W_{t+1} - W_t \rangle_F + \frac{L}{2}\|W_{t+1} - W_t\|_F^2$$
$$= f(W_t) - \eta \langle \nabla f(W_t), \widetilde{O}_t \rangle_F + \frac{\eta^2 L}{2}\|\widetilde{O}_t\|_F^2.$$

We first analyze the quadratic term $\|\widetilde{O}_t\|_F^2$. For the $m \leq n$ regime, the geometry-aligned adaptive update $\widetilde{O}_t$ is given by $\widetilde{O}_t = \text{Diag}(s_t)O_t$. Since $O_t = U_t V_t^\top$ is the polar factor of M_t, we have $O_t O_t^\top = U_t V_t^\top V_t U_t^\top = U_t U_t^\top = I_m$. Then using the definition of the Frobenius norm and the cyclic property of the trace, we have

$$\|\widetilde{O}_t\|_F^2 = \text{tr}(\widetilde{O}_t^\top \widetilde{O}_t) = \text{tr}\left(O_t^\top \text{Diag}(s_t)^2 O_t\right) = \text{tr}\left(\text{Diag}(s_t)^2 O_t O_t^\top\right) = \text{tr}\left(\text{Diag}(s_t)^2\right) = \sum_{i=1}^m s_{t,i}^2.$$

Let $s_{t,\max} = \max_i s_{t,i}$ denote the maximum entry of the scaling vector s_t, then we have

$$\|\widetilde{O}_t\|_F^2 = \sum_{i=1}^m s_{t,i}^2 \leq m s_{t,\max}^2.$$

For the $m > n$ regime, the adaptive update is given by $\widetilde{O}_t = O_t \,\text{Diag}(s_t)$. By definition of the polar factor, we have $O_t^\top O_t = I_n$. Applying an analogous derivation, we obtain $\|\widetilde{O}_t\|_F^2 = \sum_{j=1}^n s_{t,j}^2 \leq n s_{t,\max}^2$. By introducing $\ell = \min(m, n)$, both regimes can be unified and compactly written as $\|\widetilde{O}_t\|_F^2 \leq \ell s_{t,\max}^2$. Consequently, the quadratic term in the smoothness inequality satisfies

$$\frac{\eta^2 L}{2}\|\widetilde{O}_t\|_F^2 = \frac{\eta^2 L}{2}\sum_{k=1}^{\ell} s_{t,k}^2 \leq \frac{\eta^2 \ell L s_{t,\max}^2}{2}.$$

We now analyze the Frobenius inner product term $\langle \nabla f(W_t), \widetilde{O}_t \rangle_F$. By decomposing $\nabla f(W_t)$ into M_t and the residual $\nabla f(W_t) - M_t$, we obtain

$$\langle \nabla f(W_t), \widetilde{O}_t \rangle_F = \langle M_t, \widetilde{O}_t \rangle_F + \langle \nabla f(W_t) - M_t, \widetilde{O}_t \rangle_F.$$

The first term characterizes the alignment of the adaptively scaled orthogonal direction with the momentum, and the second term captures the momentum tracking error due to the discrepancy between M_t and the current gradient. We handle these two terms separately in the following. To begin, we address the first term, for which we have the following lemma.

我们根据缩放因子的中位数（用 $s_{\text{med}} = \text{median}(s_t)$ 表示）自适应地确定裁剪边界 $s_{t,\min}$ 和 $s_{t,\max}$。具体来说，我们设置 $s_{t,\min} = c \cdot s_{\text{med}}$ 和 $s_{t,\max} = d \cdot s_{\text{med}}$，其中 $c \in (0, 1]$ 和 $d \geq 1$ 控制允许的动态范围。也可以通过设置 $d = 1/c$使用单个超参数。裁剪操作有效地限制了诱导对角线缩放矩阵的条件数，并防止沿特定正交方向产生不成比例的大或小更新。当 $s_{t,\min} = s_{t,\max}$时，裁剪会移除所有相对自适应性，将更新简化为均匀缩放的 Muon 更新。

5 理论分析

在本节中，我们建立了DGA-Muon在精确正交化下的收敛保证。我们首先介绍分析所需的假设。

假设。我们在整个分析中做如下假设。

(A1) (L-平滑性) $\|\nabla f(X) - \nabla f(Y)\|_F \leq L\|X - Y\|_F$，对于所有 X, Y。

(A2) (具有有界方差的非偏差随机梯度) 随机梯度满足

$$\mathbb{E}[G_t \mid \mathcal{F}_{t-1}] = \nabla f(W_t), \quad \mathbb{E}\|G_t - \nabla f(W_t)\|_F^2 \leq \sigma^2.$$

(A3) (有界随机梯度)存在一个常数 $G_{\max} > 0$，使得

$$\|G_t\|_F \leq G_{\max} \text{ for all } t.$$

我们现在进行收敛性分析。根据 f 的 L-平滑性和更新规则 $\sim W_{t+1} = W_t - \eta O_t$，我们有

$$f(W_{t+1}) \leq f(W_t) + \langle \nabla f(W_t), W_{t+1} - W_t \rangle_F + \frac{L}{2}\|W_{t+1} - W_t\|_F^2$$
$$= f(W_t) - \eta \langle \nabla f(W_t), \widetilde{O}_t \rangle_F + \frac{\eta^2 L}{2}\|\widetilde{O}_t\|_F^2.$$

我们首先分析二次项 $\|\widetilde{O}_t\|_F^2$。对于 $m \leq n$ 情形，几何对齐的自适应更新 $\widetilde{O}_t$ 由 $\widetilde{O}_t = \text{Diag}(s_t) O_t$ 给出。由于 $O_t = U_t V_t^\top$ 是 M_t 的极因子，我们有 $O_t O_t^\top = U_t V_t^\top V_t U_t^\top = U_t U_t^\top = I_m$。然后利用 Frobenius 范数的定义和迹的循环性质，我们有

$$\|\widetilde{O}_t\|_F^2 = \text{tr}(\widetilde{O}_t^\top \widetilde{O}_t) = \text{tr}\left(O_t^\top \text{Diag}(s_t)^2 O_t\right) = \text{tr}\left(\text{Diag}(s_t)^2 O_t O_t^\top\right) = \text{tr}\left(\text{Diag}(s_t)^2\right) = \sum_{i=1}^m s_{t,i}^2.$$

令 $s_{t,\max} = \max_i s_{t,i}$ 表示缩放向量 s_t 的最大分量，则我们有

$$\|\widetilde{O}_t\|_F^2 = \sum_{i=1}^m s_{t,i}^2 \leq m s_{t,\max}^2.$$

~对于 $m > n$ 情形，自适应更新由 $O_t = O_t \text{Diag}(s_t)$ 给出。根据极因子的定义，我们有 $O_t^\top O_t = I_n$。应用类似的推导，我们得到 $\|\widetilde{O}_t\|_F^2 = \sum_{j=1}^n s_{t,j}^2 \leq n s_{t,\max}^2$。通过引入 $\ell = \min(m, n)$，两种情形可以统一并紧凑地表示为 $\|\widetilde{O}_t\|_F^2 \leq \ell s_{t,\max}^2$。因此，平滑性不等式中的二次项满足

$$\frac{\eta^2 L}{2}\|\widetilde{O}_t\|_F^2 = \frac{\eta^2 L}{2}\sum_{k=1}^{\ell} s_{t,k}^2 \leq \frac{\eta^2 \ell L s_{t,\max}^2}{2}.$$

~我们现在分析Frobenius内积项 $\langle \nabla f(W_t)\ O_t \rangle_F$.通过将 $\nabla f(W_t)$ 分解为 M_t 和残差 $\nabla f(W_t) - M_t$，我们得到

$$\langle \nabla f(W_t), \widetilde{O}_t \rangle_F = \langle M_t, \widetilde{O}_t \rangle_F + \langle \nabla f(W_t) - M_t, \widetilde{O}_t \rangle_F.$$

第一个项表征自适应缩放的正交方向与动量的对齐，第二个项则捕获由于 M_t 与当前梯度之间的差异而产生的动量跟踪误差。我们在以下内容中分别处理这两个项。首先，我们处理第一个项，该项有如下引理。

Lemma 1 (Adaptive Polar Alignment Lower Bound)**.** *Let M_t, O_t, $\widetilde{O}_t$ and s_t be defined as in Algorithm 1. Let $s_{t,\min} = \min_i s_{t,i}$ denote the minimum entry of the scaling vector s_t. Then, the following lower bound on the adaptive polar alignment holds*

$$\langle M_t, \widetilde{O}_t \rangle_F \geq s_{t,\min} \|M_t\|_F.$$

Proof. We first consider the $m \leq n$ regime, where the adaptive scaling is applied via left multiplication: $\widetilde{O}_t = \mathrm{Diag}(s_t)O_t$. Substituting $M_t = U_t \Sigma_t V_t^\top$ and $\widetilde{O}_t = \mathrm{Diag}(s_t)U_t V_t^\top$ into the Frobenius inner product $\langle M_t, \widetilde{O}_t \rangle_F = \mathrm{tr}(M_t^\top \widetilde{O}_t)$, we obtain

$$\langle M_t, \widetilde{O}_t \rangle_F = \mathrm{tr}((U_t \Sigma_t V_t^\top)^\top (\mathrm{Diag}(s_t) U_t V_t^\top)) = \mathrm{tr}(V_t \Sigma_t U_t^\top \mathrm{Diag}(s_t) U_t V_t^\top).$$

Using the cyclic property of the trace and the fact that V_t has orthonormal columns (i.e., $V_t^\top V_t = I_m$), and expanding the trace, we have

$$\langle M_t, \widetilde{O}_t \rangle_F = \mathrm{tr}(\Sigma_t U_t^\top \mathrm{Diag}(s_t) U_t) = \sum_{j=1}^{m} \sigma_j \, (U_t^\top \mathrm{Diag}(s_t) U_t)_{jj},$$

where $\{\sigma_j\}_{j=1}^m$ are the singular values of M_t. By definition, the diagonal entries of $U_t^\top \mathrm{Diag}(s_t) U_t$ are given by

$$(U_t^\top \mathrm{Diag}(s_t) U_t)_{jj} = \sum_{i=1}^{m} s_{t,i} \, U_{t,ij}^2,$$

where $\{s_{t,i}\}_{i=1}^m$ are the entries of s_t. Recall that $s_{t,\min}$ denotes the minimum scaling factor of s_t, i.e., $s_{t,\min} = \min_i s_{t,i}$. Then, for each j, we have

$$\left(U_t^\top \mathrm{Diag}(s_t) U_t\right)_{jj} = \sum_{i=1}^{m} s_{t,i} U_{t,ij}^2 \geq \sum_{i=1}^{m} s_{t,\min} U_{t,ij}^2 = s_{t,\min} \sum_{i=1}^{m} U_{t,ij}^2 = s_{t,\min},$$

where the last equality follows from the column orthonormality of U_t, i.e., $U_t^\top U_t = I_m$. Therefore, we obtain a lower bound for $\langle M_t, \widetilde{O}_t \rangle_F$

$$\langle M_t, \widetilde{O}_t \rangle_F = \sum_{j=1}^{m} \sigma_j (U_t^\top \mathrm{Diag}(s_t) U_t)_{jj} \geq s_{t,\min} \sum_{j=1}^{m} \sigma_j = s_{t,\min} \|M_t\|_*,$$

where $\|\cdot\|_*$ denotes the nuclear norm of a matrix. Finally, since the norm inequality $\|A\|_* \geq \|A\|_F$ holds for any matrix A (Horn & Johnson, 2013), it follows that

$$\langle M_t, \widetilde{O}_t \rangle_F \geq s_{t,\min} \|M_t\|_F.$$

The $m > n$ regime is analogous, where the adaptive scaling is applied via right multiplication as $\widetilde{O}_t = O_t \, \mathrm{Diag}(s_t)$. Utilizing the column orthonormality of both U_t and V_t, and following an analogous argument, we obtain the same bound. Combining the two regimes, we prove the claimed bound in the lemma. □

Now, by decomposing M_t into the sum of the gradient $\nabla f(W_t)$ and the momentum tracking error $M_t - \nabla f(W_t)$, and applying the reverse triangle inequality for the Frobenius norm, we have

$$\|M_t\|_F = \|\nabla f(W_t) + (M_t - \nabla f(W_t))\|_F \geq \|\nabla f(W_t)\|_F - \|M_t - \nabla f(W_t)\|_F.$$

Combining this inequality with Lemma 1, we obtain

$$\langle M_t, \widetilde{O}_t \rangle_F \geq s_{t,\min} \|M_t\|_F \geq s_{t,\min} \|\nabla f(W_t)\|_F - s_{t,\min} \|M_t - \nabla f(W_t)\|_F.$$

We now proceed to address the term $\langle \nabla f(W_t) - M_t, \widetilde{O}_t \rangle_F$. We first consider its negation. By the Cauchy-Schwarz inequality, we have

$$\langle M_t - \nabla f(W_t), \widetilde{O}_t \rangle_F \leq \|M_t - \nabla f(W_t)\|_F \|\widetilde{O}_t\|_F.$$

As established earlier, we have $\|\widetilde{O}_t\|_F^2 \leq \ell s_{t,\max}^2$ with $\ell = \min(m, n)$. Consequently, we obtain

$$\langle M_t - \nabla f(W_t), \widetilde{O}_t \rangle_F \leq \sqrt{\ell} \|M_t - \nabla f(W_t)\|_F \, s_{t,\max}.$$

引理1 (自适应极对齐下界)。令 M_t、O_t、O_t 和 s_t 如算法*1*中定义。令 $s_{t,\min} = \min_i s_{t,i}$ 表示缩放向量 s_t 的最小分量。那么，自适应极对齐的下界如下

$$\langle M_t, \widetilde{O}_t \rangle_F \geq s_{t,\min} \|M_t\|_F.$$

证明。我们首先考虑 $m \leq n$ 情形，其中自适应缩放通过左乘应用：$\widetilde{O}_t = \mathrm{Diag}(s_t)O_t$。将 $M_t = U_t \Sigma_t V_t^\top$ 和 $\widetilde{O}_t = \mathrm{Diag}(s_t)U_t V_t^\top$ 代入 Frobenius 内积 $\langle M_t, \widetilde{O}_t \rangle_F = \mathrm{tr}(M_t^\top \widetilde{O}_t)$，我们得到

$$\langle M_t, \widetilde{O}_t \rangle_F = \mathrm{tr}((U_t \Sigma_t V_t^\top)^\top (\mathrm{Diag}(s_t) U_t V_t^\top)) = \mathrm{tr}(V_t \Sigma_t U_t^\top \mathrm{Diag}(s_t) U_t V_t^\top).$$

Usi利用迹的循环性质以及 V_t 具有正交归一列（即 $V_t^\top V_t =$）这一事实 I_m)，并展开迹，我们有

$$\langle M_t, \widetilde{O}_t \rangle_F = \mathrm{tr}(\Sigma_t U_t^\top \mathrm{Diag}(s_t) U_t) = \sum_{j=1}^{m} \sigma_j \, (U_t^\top \mathrm{Diag}(s_t) U_t)_{jj},$$

其中 $\{\sigma_j\}_{j=1}^m$ 是 M_t 的奇异值。根据定义， $U_t^\top \mathrm{Diag}(s_t) U_t$ 的对角线元素由下式给出

$$(U_t^\top \mathrm{Diag}(s_t) U_t)_{jj} = \sum_{i=1}^{m} s_{t,i} \, U_{t,ij}^2,$$

其中 $\{s_{t,i}\}_{i=1}^m$ 是 s_t 的元素。回想一下，$s_{t,\min}$ 表示 s_t 的最小缩放因子，即 $s_{t,\min} = \min_i s_{t,i}$. 然后，对于每个 j，我们有

$$\left(U_t^\top \mathrm{Diag}(s_t) U_t\right)_{jj} = \sum_{i=1}^{m} s_{t,i} U_{t,ij}^2 \geq \sum_{i=1}^{m} s_{t,\min} U_{t,ij}^2 = s_{t,\min} \sum_{i=1}^{m} U_{t,ij}^2 = s_{t,\min},$$

其中最后一个等式源自 U_t 的列正交规范性，即 $U_t^\top U_t = I_m$。因此，我们得到 $\langle M_t, O_t \rangle_F$ 的下界

$$\langle M_t, \widetilde{O}_t \rangle_F = \sum_{j=1}^{m} \sigma_j (U_t^\top \mathrm{Diag}(s_t) U_t)_{jj} \geq s_{t,\min} \sum_{j=1}^{m} \sigma_j = s_{t,\min} \|M_t\|_*,$$

其中 $\|\cdot\|_*$ 表示矩阵的核范数。最后，由于范数不等式 $\|A\|_* \geq \|A\|_F$ 对任何矩阵 A （Horn & Johnson, 2013）都成立，因此可得

$$\langle M_t, \widetilde{O}_t \rangle_F \geq s_{t,\min} \|M_t\|_F.$$

该 $m > n$ 情形类似，其中自适应缩放通过右乘 $\widetilde{}$ 作为 $O_t = O_t \mathrm{Diag}(s_t)$ 应用。利用 U_t 和 V_t 的列正交规范性，并遵循类似的论证，我们得到相同的界。结合这两种情形，我们在引理中证明了所声称的界。 □

现在，通过将 M_t 分解为梯度 $\nabla f(W_t)$ 和动量跟踪误差 $M_t - \nabla f(W_t)$ 的和，并应用 Frobenius 范数的反向三角不等式，我们有

$$\|M_t\|_F = \|\nabla f(W_t) + (M_t - \nabla f(W_t))\|_F \geq \|\nabla f(W_t)\|_F - \|M_t - \nabla f(W_t)\|_F.$$

将这个不等式与引理1结合，我们得到

$$\langle M_t, \widetilde{O}_t \rangle_F \geq s_{t,\min} \|M_t\|_F \geq s_{t,\min} \|\nabla f(W_t)\|_F - s_{t,\min} \|M_t - \nabla f(W_t)\|_F.$$

$\widetilde{}$我们现在来处理项 $\langle \nabla f(W_t) - M_t, O_t \rangle_F$。我们首先考虑它的否定。根据柯西-施瓦茨不等式，我们有

$$\langle M_t - \nabla f(W_t), \widetilde{O}_t \rangle_F \leq \|M_t - \nabla f(W_t)\|_F \|\widetilde{O}_t\|_F.$$

如前所述，我们已有 $\|\widetilde{O}_t\|_F^2 \leq \ell s_{t,\ \max}^2$ 与 $\ell = \min(m,\ n)$。因此，我们得到

$$\langle M_t - \nabla f(W_t), \widetilde{O}_t \rangle_F \leq \sqrt{\ell} \|M_t - \nabla f(W_t)\|_F \, s_{t,\max}.$$

Therefore, by negating the above inequality, we obtain

$$\langle \nabla f(W_t) - M_t, \widetilde{O}_t \rangle_F \geq -\sqrt{\ell}\|M_t - \nabla f(W_t)\|_F s_{t,\max}.$$

Substituting the above bounds for the alignment and tracking error terms into the smoothness inequality, we obtain the following descent lemma for DGA-Muon.

Lemma 2 (Per-step Descent Inequality)**.** *Let W_t, M_t and s_t be defined as in Algorithm 1. Let $s_{t,\min}$ and $s_{t,\max}$ denote the minimum and maximum entries of s_t, respectively, and let $\ell = \min(m, n)$. Assume that f is L-smooth. Then, the updates of DGA-Muon satisfy the following per-step descent inequality*

$$f(W_{t+1}) \leq f(W_t) - \eta s_{t,\min}\|\nabla f(W_t)\|_F + \eta(s_{t,\min} + \sqrt{\ell} s_{t,\max})\|M_t - \nabla f(W_t)\|_F + \frac{\eta^2 \ell L s_{t,\max}^2}{2}.$$

Next, we analyze $\|M_t - \nabla f(W_t)\|_F$, which quantifies the deviation between the stochastic momentum and the true gradient. Recall that $M_t = \beta_1 M_{t-1} + (1-\beta_1)G_t$. Subtracting the true gradient $\nabla f(W_t)$, we have

$$M_t - \nabla f(W_t) = \beta_1 M_{t-1} + (1-\beta_1)G_t - \nabla f(W_t).$$

To isolate the previous-step error, we add and subtract $\beta_1 \nabla f(W_{t-1})$

$$M_t - \nabla f(W_t) = \beta_1(M_{t-1} - \nabla f(W_{t-1})) + (1-\beta_1)(G_t - \nabla f(W_t)) + \beta_1(\nabla f(W_{t-1}) - \nabla f(W_t)).$$

Unrolling recursively from t down to 1 gives

$$M_t - \nabla f(W_t) = \beta_1^{t-1}(M_1 - \nabla f(W_1)) + (1-\beta_1)\sum_{s=2}^{t} \beta_1^{t-s}(G_s - \nabla f(W_s))$$
$$+ \beta_1 \sum_{s=2}^{t} \beta_1^{t-s}(\nabla f(W_{s-1}) - \nabla f(W_s)).$$

Taking the Frobenius norm and applying the triangle inequality, we obtain

$$\|M_t - \nabla f(W_t)\|_F \leq \beta_1^{t-1}\|M_1 - \nabla f(W_1)\|_F + \Big\|(1-\beta_1)\sum_{s=2}^{t} \beta_1^{t-s}(G_s - \nabla f(W_s))\Big\|_F$$
$$+ \beta_1 \sum_{s=2}^{t} \beta_1^{t-s}\|\nabla f(W_{s-1}) - \nabla f(W_s)\|_F.$$

The right-hand side naturally decomposes into three contributions: the initialization term, the stochastic noise term, and the gradient drift term. In the following, we analyze each of them separately.

For the initialization term, since $M_1 = (1-\beta_1)G_1$, we decompose

$$M_1 - \nabla f(W_1) = (1-\beta_1)(G_1 - \nabla f(W_1)) - \beta_1 \nabla f(W_1).$$

Therefore,

$$\mathbb{E}\|\beta_1^{t-1}(M_1 - \nabla f(W_1))\|_F \leq \beta_1^{t-1}(1-\beta_1)\mathbb{E}\|G_1 - \nabla f(W_1)\|_F + \beta_1^t\|\nabla f(W_1)\|_F$$

Using the unbiasedness of G_1, Jensen's inequality and the gradient bound assumption, we have

$$\|\nabla f(W_1)\|_F = \|\mathbb{E}[G_1|\mathcal{F}_0]\|_F \leq \mathbb{E}[\|G_1\|_F|\mathcal{F}_0] \leq G_{\max}.$$

Then applying Jensen's inequality, $\mathbb{E}\|X\|_F \leq \sqrt{\mathbb{E}\|X\|_F^2}$, and using the bounded-variance assumption $\mathbb{E}\|G_1 - \nabla f(W_1)\|_F^2 \leq \sigma^2$, we obtain

$$\mathbb{E}\big[\|\beta_1^{t-1}(M_1 - \nabla f(W_1))\|_F\big] \leq \beta_1^{t-1}(1-\beta_1)\sigma + \beta_1^t G_{\max}.$$

For the stochastic noise term, we first bound its second moment. For notational convenience, we introduce the shorthand $\mathcal{N}_t = (1-\beta_1)\sum_{s=2}^{t} \beta_1^{t-s}(G_s - \nabla f(W_s))$. Expanding the squared Frobenius norm, taking the expectation, and using the linearity of expectation, we obtain

$$\mathbb{E}\|\mathcal{N}_t\|_F^2 = (1-\beta_1)^2\,\mathbb{E}\Big\langle \sum_{s=2}^{t} \beta_1^{t-s}(G_s - \nabla f(W_s)),\ \sum_{r=2}^{t} \beta_1^{t-r}(G_r - \nabla f(W_r))\Big\rangle$$
$$= (1-\beta_1)^2 \sum_{s=2}^{t}\sum_{r=2}^{t} \beta_1^{2t-s-r}\,\mathbb{E}\big\langle G_s - \nabla f(W_s),\ G_r - \nabla f(W_r)\big\rangle.$$

因此，通过否定上述不等式，我们得到

$$\langle \nabla f(W_t) - M_t, \widetilde{O}_t \rangle_F \geq -\sqrt{\ell}\|M_t - \nabla f(W_t)\|_F s_{t,\max}.$$

将上述对齐和跟踪误差项的界代入平滑性不等式，我们得到 DGA-Muon 的以下下降引理。

引理2(每步下降不等式)。 令 W_t, M_t 和 s_t 如算法*1*中定义。令 $s_{t,\min}$ 和 $s_{t,\max}$ 分别表示 s_t 的最小值和最大值，并令 $\ell = \min(m, n)$。假设 f 是 L- 平滑的。那么，*DGA-Muon* 的更新满足以下每步下降不等式

$$f(W_{t+1}) \leq f(W_t) - \eta s_{t,\min}\|\nabla f(W_t)\|_F + \eta(s_{t,\min} + \sqrt{\ell} s_{t,\max})\|M_t - \nabla f(W_t)\|_F + \frac{\eta^2 \ell L s_{t,\max}^2}{2}.$$

接下来，我们分析 $\|M_t - \nabla f(W_t)\|_F$，它量化了随机动量与真实梯度之间的偏差。回想 $M_t = \beta_1 M_{t-1} + (1-\beta_1)G_t$。减去真实梯度 $\nabla f(W_t)$，我们得到

$$M_t - \nabla f(W_t) = \beta_1 M_{t-1} + (1-\beta_1)G_t - \nabla f(W_t).$$

为了隔离前一步误差，我们加上并减去 $\beta_1 \nabla f(W_{t-1})$

$$M_t - \nabla f(W_t) = \beta_1(M_{t-1} - \nabla f(W_{t-1})) + (1-\beta_1)(G_t - \nabla f(W_t)) + \beta_1(\nabla f(W_{t-1}) - \nabla f(W_t)).$$

从 t递归地展开到1得到

$$M_t - \nabla f(W_t) = \beta_1^{t-1}(M_1 - \nabla f(W_1)) + (1-\beta_1)\sum_{s=2}^{t} \beta_1^{t-s}(G_s - \nabla f(W_s))$$
$$+ \beta_1 \sum_{s=2}^{t} \beta_1^{t-s}(\nabla f(W_{s-1}) - \nabla f(W_s)).$$

取Frobenius范数并应用三角不等式，我们得到

$$\|M_t - \nabla f(W_t)\|_F \leq \beta_1^{t-1}\|M_1 - \nabla f(W_1)\|_F + \Big\|(1-\beta_1)\sum_{s=2}^{t} \beta_1^{t-s}(G_s - \nabla f(W_s))\Big\|_F$$
$$+ \beta_1 \sum_{s=2}^{t} \beta_1^{t-s}\|\nabla f(W_{s-1}) - \nabla f(W_s)\|_F.$$

右侧自然分解为三项贡献：初始化项、随机噪声项和梯度漂移项。在以下内容中，我们分别分析每一项。

对于初始化项，由于 $M_1 = (1-\beta_1)G_1$，我们分解

$$M_1 - \nabla f(W_1) = (1-\beta_1)(G_1 - \nabla f(W_1)) - \beta_1 \nabla f(W_1).$$

因此，

$$\mathbb{E}\|\beta_1^{t-1}(M_1 - \nabla f(W_1))\|_F \leq \beta_1^{t-1}(1-\beta_1)\mathbb{E}\|G_1 - \nabla f(W_1)\|_F + \beta_1^t\|\nabla f(W_1)\|_F$$

利用 G_1的无偏性、Jensen不等式和梯度界假设，我们有

$$\|\nabla f(W_1)\|_F = \|\mathbb{E}[G_1|\mathcal{F}_0]\|_F \leq \mathbb{E}[\|G_1\|_F|\mathcal{F}_0] \leq G_{\max}.$$

然后应用Jensen不等式，$\mathbb{E}\|X\|_F \leq \sqrt{\mathbb{E}\|X\|_F^2}$，并使用有界方差假设 $\mathbb{E}\|G_1 - \nabla f(W_1)\|_F^2 \leq \sigma^2$，我们得到

$$\mathbb{E}\big[\|\beta_1^{t-1}(M_1 - \nabla f(W_1))\|_F\big] \leq \beta_1^{t-1}(1-\beta_1)\sigma + \beta_1^t G_{\max}.$$

对于随机噪声项，我们首先对其二阶矩进行界。为方便记号，我们引入缩写 $\mathcal{N}_t = (1-\beta_1)\sum_{s=2}^{t} \beta_1^{t-s}(G_s - \nabla f(W_s))$。展开平方Frobenius范数，取期望，并利用期望的线性，我们得到

$$\mathbb{E}\|\mathcal{N}_t\|_F^2 = (1-\beta_1)^2\,\mathbb{E}\Big\langle \sum_{s=2}^{t} \beta_1^{t-s}(G_s - \nabla f(W_s)),\ \sum_{r=2}^{t} \beta_1^{t-r}(G_r - \nabla f(W_r))\Big\rangle$$
$$= (1-\beta_1)^2 \sum_{s=2}^{t}\sum_{r=2}^{t} \beta_1^{2t-s-r}\,\mathbb{E}\big\langle G_s - \nabla f(W_s),\ G_r - \nabla f(W_r)\big\rangle.$$

By assumption (A2), the stochastic gradient noise $\xi_t = G_t - \nabla f(W_t)$ forms a martingale difference sequence, i.e., $\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$, where $\mathcal{F}_{t-1}$ denotes the filtration representing all information up to time $t-1$. Hence, the cross terms ($s \neq r$) vanish in expectation, leaving only the diagonal terms

$$\mathbb{E}\|\mathcal{N}_t\|_F^2 = (1-\beta_1)^2 \sum_{s=2}^{t} \beta_1^{2(t-s)}\, \mathbb{E}\|G_s - \nabla f(W_s)\|_F^2 \le (1-\beta_1)^2\sigma^2 \sum_{k=0}^{t-2} \beta_1^{2k}.$$

Evaluating the geometric sum gives

$$\mathbb{E}\|\mathcal{N}_t\|_F^2 \le (1-\beta_1)^2\sigma^2 \frac{1-\beta_1^{2(t-1)}}{1-\beta_1^2} = \sigma^2 \frac{(1-\beta_1)(1-\beta_1^{2(t-1)})}{1+\beta_1} \le \sigma^2 \frac{1-\beta_1}{1+\beta_1}.$$

Then applying Jensen's inequality and using the bounded-variance assumption, we obtain

$$\mathbb{E}\|\mathcal{N}_t\|_F \le \sqrt{\mathbb{E}\|\mathcal{N}_t\|_F^2} \le \frac{\sqrt{1-\beta_1}}{\sqrt{1+\beta_1}}\sigma.$$

We now proceed to handle the gradient drift term, which, unlike the stochastic noise term, can be bounded deterministically. By the L-smoothness of f, the update rule $W_s = W_{s-1} - \eta\widetilde{O}_{s-1}$, and the bound $\|\widetilde{O}_t\|_F \le \sqrt{\ell}\, s_{t,\max}$, we have

$$\begin{aligned}\beta_1 \sum_{s=2}^{t} \beta_1^{t-s}\|\nabla f(W_{s-1}) - \nabla f(W_s)\|_F &\le \beta_1 \sum_{s=2}^{t} \beta_1^{t-s} L\|W_{s-1} - W_s\|_F \\ &= \beta_1\eta L \sum_{s=2}^{t} \beta_1^{t-s}\|\widetilde{O}_{s-1}\|_F \le \beta_1\eta\sqrt{\ell}L \sum_{s=2}^{t} \beta_1^{t-s} s_{s-1,\max}.\end{aligned}$$

Let $s_{\max}$ be the maximum of all $s_{t,\max}$, then we obtain

$$\beta_1 \sum_{s=2}^{t} \beta_1^{t-s}\|\nabla f(W_{s-1}) - \nabla f(W_s)\|_F \le \beta_1\eta\sqrt{\ell}L s_{\max} \sum_{s=2}^{t} \beta_1^{t-s}$$

For the geometric sum, we have

$$\sum_{s=2}^{t} \beta_1^{t-s} = \sum_{k=0}^{t-2} \beta_1^{k} = \frac{1-\beta_1^{t-1}}{1-\beta_1} \le \frac{1}{1-\beta_1}.$$

Therefore, the gradient drift term admits the following deterministic bound

$$\beta_1 \sum_{s=2}^{t} \beta_1^{t-s}\|\nabla f(W_{s-1}) - \nabla f(W_s)\|_F \le \frac{\beta_1\eta\sqrt{\ell}L s_{\max}}{1-\beta_1}.$$

Combining the bounds for these three terms yields the following bound on the expected momentum error.

Lemma 3 (Expected Momentum Error Bound)**.** *Let M_t and W_t be defined as in Algorithm 1. Let $s_{\max}$ denote a uniform upper bound on the adaptive scaling factors satisfying $(s_t)_i \le s_{\max} < \infty$ for all t and i, and let $\ell = \min(m,n)$. Assume that f is L-smooth. Then, under the bounded-variance and bounded-gradient assumptions, the expected momentum error at each step satisfies*

$$\mathbb{E}\|M_t - \nabla f(W_t)\|_F \le \beta_1^{t-1}(1-\beta_1)\sigma + \beta_1^t G_{\max} + \frac{\sqrt{1-\beta_1}}{\sqrt{1+\beta_1}}\sigma + \frac{\beta_1\eta\sqrt{\ell}L s_{\max}}{1-\beta_1}.$$

Recall that Lemma 2 establishes the following one-step descent inequality for the updates of Algorithm 1

$$f(W_{t+1}) \le f(W_t) - \eta s_{t,\min}\|\nabla f(W_t)\|_F + \eta(s_{t,\min} + \sqrt{\ell}\, s_{t,\max})\|M_t - \nabla f(W_t)\|_F + \frac{\eta^2\ell L s_{t,\max}^2}{2}.$$

To simplify the subsequent analysis, we replace the time-dependent quantities by uniform bounds over the optimization horizon. Let $s_{\min}$ denote a uniform lower bound on the adaptive scaling factors satisfying $0 < s_{\min} \le (s_t)_i$ for all t and i. Since

$$s_{\min} \le s_{t,\min} \le s_{t,\max} \le s_{\max},$$

根据假设（A2），随机梯度噪声 $\xi_t = G_t - \nabla f(W_t)$ 形成一个鞅差序列，即 $\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$，其中 $\mathcal{F}_{t-1}$ 表示代表直到时间 $t-1$ 的所有信息的滤波。因此，交叉项 ($s \neq r$) 在期望中消失，只留下对角项

$$\mathbb{E}\|\mathcal{N}_t\|_F^2 = (1-\beta_1)^2 \sum_{s=2}^{t} \beta_1^{2(t-s)}\, \mathbb{E}\|G_s - \nabla f(W_s)\|_F^2 \le (1-\beta_1)^2\sigma^2 \sum_{k=0}^{t-2} \beta_1^{2k}.$$

计算几何和得到

$$\mathbb{E}\|\mathcal{N}_t\|_F^2 \le (1-\beta_1)^2\sigma^2 \frac{1-\beta_1^{2(t-1)}}{1-\beta_1^2} = \sigma^2 \frac{(1-\beta_1)(1-\beta_1^{2(t-1)})}{1+\beta_1} \le \sigma^2 \frac{1-\beta_1}{1+\beta_1}.$$

然后应用Jensen不等式并使用有界方差假设，我们得到

$$\mathbb{E}\|\mathcal{N}_t\|_F \le \sqrt{\mathbb{E}\|\mathcal{N}_t\|_F^2} \le \frac{\sqrt{1-\beta_1}}{\sqrt{1+\beta_1}}\sigma.$$

我们现在处理梯度漂移项，它与随机噪声项不同，可以被~确定性有界。根据 f的 L-平滑性、更新规则 $W_s = W_{s-1} - \eta O_{s-1}$以及界$\|\widetilde{O}_t\|_F \le \sqrt{\ell}\, s_{t,\max}$，我们有

$$\begin{aligned}\beta_1 \sum_{s=2}^{t} \beta_1^{t-s}\|\nabla f(W_{s-1}) - \nabla f(W_s)\|_F &\le \beta_1 \sum_{s=2}^{t} \beta_1^{t-s} L\|W_{s-1} - W_s\|_F \\ &= \beta_1\eta L \sum_{s=2}^{t} \beta_1^{t-s}\|\widetilde{O}_{s-1}\|_F \le \beta_1\eta\sqrt{\ell}L \sum_{s=2}^{t} \beta_1^{t-s} s_{s-1,\max}.\end{aligned}$$

设 $s_{\max}$ 是所有$s_{t,\max}$的最大值，则我们得到

$$\beta_1 \sum_{s=2}^{t} \beta_1^{t-s}\|\nabla f(W_{s-1}) - \nabla f(W_s)\|_F \le \beta_1\eta\sqrt{\ell}L s_{\max} \sum_{s=2}^{t} \beta_1^{t-s}$$

对于几何和，我们有

$$\sum_{s=2}^{t} \beta_1^{t-s} = \sum_{k=0}^{t-2} \beta_1^{k} = \frac{1-\beta_1^{t-1}}{1-\beta_1} \le \frac{1}{1-\beta_1}.$$

因此，梯度漂移项具有以下确定性界

$$\beta_1 \sum_{s=2}^{t} \beta_1^{t-s}\|\nabla f(W_{s-1}) - \nabla f(W_s)\|_F \le \frac{\beta_1\eta\sqrt{\ell}L s_{\max}}{1-\beta_1}.$$

结合这三个项的界，得到期望动量误差的如下界。

引理3(期望动量误差界)。 令 M_t 和 W_t 按算法*1*定义。令$s_{\max}$ 表示自适应缩放因子的统一上界，满足 $(s_t)_i \le s_{\max} < \infty$对所有 t 和 i成立，并令 $\ell = \min(m,n)$。假设 f 是 L-平滑的。那么，在有界方差和有界梯度的假设下，每一步的期望动量误差满足

$$\mathbb{E}\|M_t - \nabla f(W_t)\|_F \le \beta_1^{t-1}(1-\beta_1)\sigma + \beta_1^t G_{\max} + \frac{\sqrt{1-\beta_1}}{\sqrt{1+\beta_1}}\sigma + \frac{\beta_1\eta\sqrt{\ell}L s_{\max}}{1-\beta_1}.$$

回想一下，引理2为算法1的更新建立了如下单步下降不等式

$$f(W_{t+1}) \le f(W_t) - \eta s_{t,\min}\|\nabla f(W_t)\|_F + \eta(s_{t,\min} + \sqrt{\ell}\, s_{t,\max})\|M_t - \nabla f(W_t)\|_F + \frac{\eta^2\ell L s_{t,\max}^2}{2}.$$

为简化后续分析，我们将时间相关的量替换为在优化范围内均匀的界。令 $s_{\min}$ 表示自适应缩放因子的均匀下界，它满足 $0 < s_{\min} \le (s_t)_i$ 对所有 t 和 i。由于

$$s_{\min} \le s_{t,\min} \le s_{t,\max} \le s_{\max},$$

we have

$$-\eta s_{t,\min}\|\nabla f(W_t)\|_F \le -\eta s_{\min}\|\nabla f(W_t)\|_F,$$

and

$$s_{t,\min} + \sqrt{\ell}\, s_{t,\max} \le (1+\sqrt{\ell}) s_{\max}.$$

Moreover, $s_{t,\max}^2 \le s_{\max}^2$. Therefore, the one-step descent inequality can be uniformly bounded as

$$f(W_{t+1}) \le f(W_t) - \eta s_{\min}\|\nabla f(W_t)\|_F + \eta(1+\sqrt{\ell}) s_{\max}\|M_t - \nabla f(W_t)\|_F + \frac{\eta^2 \ell L s_{\max}^2}{2}.$$

Rearranging to isolate the gradient term and taking expectations on both sides yields

$$\eta s_{\min}\mathbb{E}\big[\|\nabla f(W_t)\|_F\big] \le \mathbb{E}[f(W_t) - f(W_{t+1})] + \eta(1+\sqrt{\ell}) s_{\max}\mathbb{E}\big[\|M_t - \nabla f(W_t)\|_F\big] + \frac{\eta^2 \ell L s_{\max}^2}{2}.$$

Then, summing over $t = 1, \ldots, T$, using the telescoping sum $\sum_{t=1}^T \mathbb{E}[f(W_t) - f(W_{t+1})] = \mathbb{E}[f(W_1) - f(W_{T+1})]$, and subsequently dividing by $\eta s_{\min}$, we obtain

$$\sum_{t=1}^T \mathbb{E}[\|\nabla f(W_t)\|_F] \le \frac{\mathbb{E}[f(W_1) - f(W_{T+1})]}{\eta s_{\min}} + \frac{(1+\sqrt{\ell}) s_{\max}}{s_{\min}} \sum_{t=1}^T \mathbb{E}[\|M_t - \nabla f(W_t)\|_F] + \frac{\eta \ell L s_{\max}^2 T}{2 s_{\min}}.$$

Summing the momentum error inequality given by Lemma 3 from $t = 1$ to T yields

$$\sum_{t=1}^T \mathbb{E}\|M_t - \nabla f(W_t)\|_F \le (1-\beta_1)\sigma \sum_{t=1}^T \beta_1^{t-1} + G_{\max} \sum_{t=1}^T \beta_1^t + T\frac{\sqrt{1-\beta_1}}{\sqrt{1+\beta_1}}\sigma + T\frac{\beta_1 \eta \sqrt{\ell} L s_{\max}}{1-\beta_1}.$$

Evaluating the geometric sums $\sum_{t=1}^T \beta_1^{t-1} \le \frac{1}{1-\beta_1}$ and $\sum_{t=1}^T \beta_1^t \le \frac{\beta_1}{1-\beta_1}$, we obtain

$$\sum_{t=1}^T \mathbb{E}\|M_t - \nabla f(W_t)\|_F \le \sigma + \frac{\beta_1 G_{\max}}{1-\beta_1} + T\frac{\sqrt{1-\beta_1}}{\sqrt{1+\beta_1}}\sigma + T\frac{\beta_1 \eta \sqrt{\ell} L s_{\max}}{1-\beta_1}.$$

Denote $\Delta = f(W_1) - f^\star$, where $f^\star = \inf_W f(W)$. Since $f(W_{T+1}) \ge f^\star$, we have

$$\mathbb{E}[f(W_1) - f(W_{T+1})] \le \Delta.$$

Substituting the bound for $\sum_{t=1}^T \mathbb{E}\|M_t - \nabla f(W_t)\|_F$ and $\mathbb{E}[f(W_1) - f(W_{T+1})] \le \Delta$ into the previous inequality, we obtain the final bound for the cumulative expected gradient norm:

$$\sum_{t=1}^T \mathbb{E}[\|\nabla f(W_t)\|_F] \le \frac{\Delta}{\eta s_{\min}} + \frac{(1+\sqrt{\ell}) s_{\max}}{s_{\min}}\left(\sigma + \frac{\beta_1 G_{\max}}{1-\beta_1}\right) + T\frac{(1+\sqrt{\ell}) s_{\max}}{s_{\min}}\frac{\sqrt{1-\beta_1}}{\sqrt{1+\beta_1}}\sigma + T\frac{\beta_1 \eta \sqrt{\ell} L (1+\sqrt{\ell}) s_{\max}^2}{(1-\beta_1) s_{\min}} + T\frac{\eta \ell L s_{\max}^2}{2 s_{\min}}.$$

Dividing both sides by T immediately yields a bound on the average expected gradient norm. Since the minimum is always upper bounded by the average, i.e., $\min_{1\le t\le T} \mathbb{E}[\|\nabla f(W_t)\|_F] \le \frac{1}{T}\sum_{t=1}^T \mathbb{E}[\|\nabla f(W_t)\|_F]$, we obtain the final convergence bound.

Theorem 1 (Convergence Guarantee of DGA-Muon)**.** *Let W_t be defined as in Algorithm 1. Let $s_{\min}$ and $s_{\max}$ denote uniform lower and upper bounds on the adaptive scaling factors satisfying $0 < s_{\min} \le (s_t)_i \le s_{\max} < \infty$ for all t and i, and let $\ell = \min(m, n)$. Denote $\Delta = f(W_1) - f^\star$. Assume that f is L-smooth and that the stochastic gradients are unbiased with bounded variance*

我们有

$$-\eta s_{t,\min}\|\nabla f(W_t)\|_F \le -\eta s_{\min}\|\nabla f(W_t)\|_F,$$

and

$$s_{t,\min} + \sqrt{\ell}\, s_{t,\max} \le (1+\sqrt{\ell}) s_{\max}.$$

More因此，单步下降不等式可以统一界定为

$$f(W_{t+1}) \le f(W_t) - \eta s_{\min}\|\nabla f(W_t)\|_F + \eta(1+\sqrt{\ell}) s_{\max}\|M_t - \nabla f(W_t)\|_F + \frac{\eta^2 \ell L s_{\max}^2}{2}.$$

重新排列以隔离梯度项，并对两边取期望得到

$$\eta s_{\min}\mathbb{E}\big[\|\nabla f(W_t)\|_F\big] \le \mathbb{E}[f(W_t) - f(W_{t+1})] + \eta(1+\sqrt{\ell}) s_{\max}\mathbb{E}\big[\|M_t - \nabla f(W_t)\|_F\big] + \frac{\eta^2 \ell L s_{\max}^2}{2}.$$

然后，对 $t = 1 \ldots T$ $\mathbb{E}[f\ W_1 - f\ W_{T+1}]$, $\eta s_{\min}$ $\sum_{t=1}^T \mathbb{E}[f\ W_t - f\ W_{t+1}] =$ ，求和，使用望远镜求和公式( ) ( ) ( ) ( )，然后除以，我们得到

$$\sum_{t=1}^T \mathbb{E}[\|\nabla f(W_t)\|_F] \le \frac{\mathbb{E}[f(W_1) - f(W_{T+1})]}{\eta s_{\min}} + \frac{(1+\sqrt{\ell}) s_{\max}}{s_{\min}} \sum_{t=1}^T \mathbb{E}[\|M_t - \nabla f(W_t)\|_F] + \frac{\eta \ell L s_{\max}^2 T}{2 s_{\min}}.$$

从 $t = 1$ 到 T 求和引理3给出的动量误差不等式得到

$$\sum_{t=1}^T \mathbb{E}\|M_t - \nabla f(W_t)\|_F \le (1-\beta_1)\sigma \sum_{t=1}^T \beta_1^{t-1} + G_{\max} \sum_{t=1}^T \beta_1^t + T\frac{\sqrt{1-\beta_1}}{\sqrt{1+\beta_1}}\sigma + T\frac{\beta_1 \eta \sqrt{\ell} L s_{\max}}{1-\beta_1}.$$

计算几何和 $\sum_{t=1}^T \beta_1^{t-1} \le \frac{1}{1-\beta_1}$ 和 $\sum_{t=1}^T \beta_1^t \le \frac{\beta_1}{1-\beta_1}$，我们得到

$$\sum_{t=1}^T \mathbb{E}\|M_t - \nabla f(W_t)\|_F \le \sigma + \frac{\beta_1 G_{\max}}{1-\beta_1} + T\frac{\sqrt{1-\beta_1}}{\sqrt{1+\beta_1}}\sigma + T\frac{\beta_1 \eta \sqrt{\ell} L s_{\max}}{1-\beta_1}.$$

记 $\Delta = f(W_1) - f^\star$，其中 $f^\star = \inf_W f(W)$。由于 $f(W_{T+1}) \ge f^\star$，因此

$$\mathbb{E}[f(W_1) - f(W_{T+1})] \le \Delta.$$

将 $\sum_{t=1}^T \mathbb{E}\|M_t - \nabla f(W_t)\|_F$ 和 $\mathbb{E}[f(W_1) - f(W_{T+1})] \le \Delta$ 的界代入前述不等式，我们得到累积期望梯度范数的最终界：

$$\sum_{t=1}^T \mathbb{E}[\|\nabla f(W_t)\|_F] \le \frac{\Delta}{\eta s_{\min}} + \frac{(1+\sqrt{\ell}) s_{\max}}{s_{\min}}\left(\sigma + \frac{\beta_1 G_{\max}}{1-\beta_1}\right) + T\frac{(1+\sqrt{\ell}) s_{\max}}{s_{\min}}\frac{\sqrt{1-\beta_1}}{\sqrt{1+\beta_1}}\sigma + T\frac{\beta_1 \eta \sqrt{\ell} L (1+\sqrt{\ell}) s_{\max}^2}{(1-\beta_1) s_{\min}} + T\frac{\eta \ell L s_{\max}^2}{2 s_{\min}}.$$

两边同时除以 T，立即得到平均期望梯度范数的界。由于最小值总是被平均值上界，即 $\min_{1\le t\le T} \mathbb{E}[\|\nabla f(W_t)\|_F] \le \frac{1}{T}\sum_{t=1}^T \mathbb{E}[\|\nabla f(W_t)\|_F]$，因此得到最终收敛界。

定理1 (DGA-Muon收敛性保证)**.** 令 W_t 按算法*1*定义。令$s_{\min}$ 和 $s_{\max}$ 表示自适应缩放因子的统一下界和上界，满足$0 < s_{\min} \le (s_t)_i \le s_{\max} < \infty$对所有 t和 i成立，并令 $\ell = \min(m, n)$。记 $\Delta = f(W_1) - f^\star$。假设 f 是 L-平滑的，且随机梯度是无偏的、方差有界

and bounded norm. Then, for a fixed step size η, DGA-Muon satisfies the following convergence guarantee

$$\min_{1\le t\le T}\mathbb{E}[\|\nabla f(W_t)\|_F] \le \frac{\Delta}{\eta s_{\min} T} + \frac{(1+\sqrt{\ell})s_{\max}}{s_{\min}T}\left(\sigma + \frac{\beta_1 G_{\max}}{1-\beta_1}\right) + \frac{(1+\sqrt{\ell})s_{\max}}{s_{\min}}\frac{\sqrt{1-\beta_1}}{\sqrt{1+\beta_1}}\sigma + \frac{\beta_1\eta\sqrt{\ell}L(1+\sqrt{\ell})s_{\max}^2}{(1-\beta_1)s_{\min}} + \frac{\eta\ell L s_{\max}^2}{2s_{\min}}.$$

In particular, this bound can be summarized as

$$\min_{1\le t\le T}\mathbb{E}[\|\nabla f(W_t)\|_F] = O\left(\frac{1}{T} + \sigma + \eta\right).$$

Therefore, for a fixed step size η, DGA-Muon achieves a convergence rate of $\mathcal{O}(1/T + \sigma + \eta)$. The adaptive scaling primarily influences the constant factors within the convergence bound. This result reveals the standard bias-variance tradeoff inherent in constant step-size stochastic optimization: the transient optimization error decays at an $\mathcal{O}(1/T)$ rate, while the stochastic gradient variance and the finite learning rate introduce an irreducible steady-state error. Consequently, DGA-Muon converges to a neighborhood of a stationary point, characterized by a bounded expected gradient norm whose radius depends on the stochastic variance σ and the step size η. Similar to classical stochastic optimization methods, a diminishing step-size schedule is expected to gradually eliminate this steady-state error, although extending the present analysis to this setting is beyond the scope of this work. Furthermore, the current analysis establishes a foundational convergence guarantee for DGA-Muon. A rigorous theoretical characterization of when the geometry-aligned adaptive scaling yields provable improvements over the original Muon update remains an open question and is left for future work.

6 Experiments

In this section, we report our experimental results. Note that, due to limited computational resources, some experiments were conducted only on a small scale, especially the ablation studies, and are not included in the current version. We will continue to expand the experimental evaluation as additional results become available.

6.1 Experimental Setup

We conduct our experiments using NanoChat (Karpathy, 2025), a highly optimized transformer pretraining framework that can serve as a testbed for evaluating optimization methods. Both the model architecture and training recipe have undergone extensive engineering and empirical tuning in the development of NanoChat, including multiple rounds of automated optimization with Autoresearch (Karpathy, 2026). As a result, NanoChat provides a strong and carefully tuned baseline for optimizer evaluation. We use the default NanoChat model architecture without modification and refer readers to the repository for full implementation details. For our experiments, we use a model that has 24 layers and 1.38B parameters in total, with full attention used across all layers. All experiments are conducted on a single node with eight NVIDIA A100 (80GB) GPUs.

We next describe the optimizer configuration used in NanoChat, which incorporates a range of advanced and carefully tuned optimization techniques. NanoChat employs the NorMuon (Li et al., 2026) optimizer. As in conventional Muon-based training, the optimizer is applied only to matrix-valued parameters, including the Q/K/V/O and FFN weight matrices, while the remaining parameters are optimized with AdamW (Loshchilov & Hutter, 2019). For orthogonalization, NanoChat uses the Polar Express (Amsel et al., 2026) technique. For the matrix-valued parameters, NanoChat employs the recently proposed cautious weight decay (Chen et al., 2026) in place of conventional decoupled weight decay. We also ablate this design choice by replacing cautious weight decay with standard decoupled weight decay and find a clear degradation in training performance.

并且范数有界。那么，对于固定的步长 η，*DGA-Muon* 满足以下收敛性保证

$$\min_{1\le t\le T}\mathbb{E}[\|\nabla f(W_t)\|_F] \le \frac{\Delta}{\eta s_{\min} T} + \frac{(1+\sqrt{\ell})s_{\max}}{s_{\min}T}\left(\sigma + \frac{\beta_1 G_{\max}}{1-\beta_1}\right) + \frac{(1+\sqrt{\ell})s_{\max}}{s_{\min}}\frac{\sqrt{1-\beta_1}}{\sqrt{1+\beta_1}}\sigma + \frac{\beta_1\eta\sqrt{\ell}L(1+\sqrt{\ell})s_{\max}^2}{(1-\beta_1)s_{\min}} + \frac{\eta\ell L s_{\max}^2}{2s_{\min}}.$$

特别是，这个界可以总结为

$$\min_{1\le t\le T}\mathbb{E}[\|\nabla f(W_t)\|_F] = O\left(\frac{1}{T} + \sigma + \eta\right).$$

因此，对于固定的步长 η，DGA-Muon 实现了收敛速度为 $\mathcal{O}(1/T + \sigma + \eta)$ 的收敛率。自适应缩放主要影响收敛界内的常数因子。这一结果揭示了常数步长随机优化中固有的标准偏差-方差权衡：瞬态优化误差以 $\mathcal{O}(1/T)$ 的速率衰减，而随机梯度方差和有限的学习率引入了不可消除的稳态误差。因此，DGA-Muon 收敛到一个以稳态点为邻域的区域，其特征是期望梯度范数有界，该范数的半径取决于随机方差 σ 和步长 η。类似于经典随机优化方法，递减的步长计划预计会逐渐消除这种稳态误差，尽管将当前分析扩展到该设置超出了本文的工作范围。此外，当前分析为 DGA-Muon 建立了基础收敛保证。当几何对齐的自适应缩放何时能比原始 Muon 更新带来可证明的改进，其严谨的理论表征仍然是一个悬而未决的问题，留待未来工作解决。

6 实验

在本节中，我们报告了实验结果。请注意，由于计算资源有限，一些实验仅在较小规模下进行，特别是消融研究，且未包含在当前版本中。我们将继续扩展实验评估，随着更多结果的可用而补充。

6.1 实验设置

我们使用 NanoChat（Karpathy，2025），一个高度优化的 Transformer 预训练框架，进行实验，该框架可作为评估优化方法的测试平台。模型架构和训练配方在 NanoChat 的开发过程中都经过了广泛的工程和经验性调优，包括多次使用 AutoResearch（Karpathy，2026）进行的自动化优化。因此，NanoChat 为优化器评估提供了一个强大且精心调优的基线。我们使用默认的 NanoChat 模型架构，未做修改，并建议读者参考仓库获取完整实现细节。在我们的实验中，我们使用一个包含 24 层和 1.38B 参数的模型，所有层都使用全注意力机制。所有实验都在单个节点上的八个 NVIDIA A100（80GB）GPU 上进行。

接下来，我们描述 NanoChat 中使用的优化器配置，该配置集成了多种先进且经过精心调整的优化技术。NanoChat 采用 NorMuon（Li 等人，2026）优化器。与传统的基于 Muon 的训练类似，优化器仅应用于矩阵值参数，包括 Q/K/V/O 和 FFN 权重矩阵，而其余参数则使用 AdamW（Loshchilov & Hutter，2019）进行优化。对于正交化，NanoChat 使用 Polar Express（Amsel 等人，2026）技术。对于矩阵值参数，NanoChat 采用最近提出的谨慎权重衰减（Chen 等人，2026），以替代传统的解耦权重衰减。我们还通过用标准解耦权重衰减替换谨慎权重衰减来消融这一设计选择，并发现训练性能明显下降。

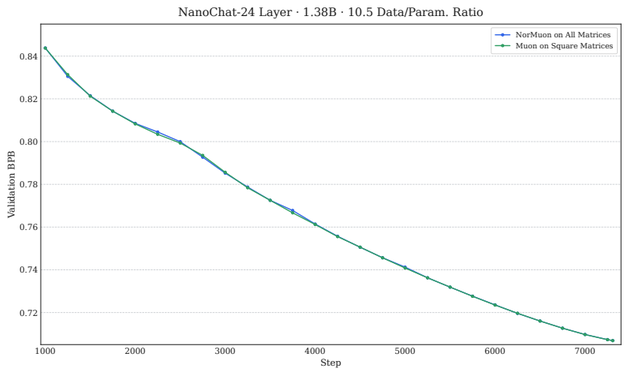


Figure 1: Empirical verification of NorMuon's adaptive scaling degeneration on square matrices in NanoChat. Removing NorMuon's adaptive scaling on the square Q/K/V/O projection matrices, effectively reducing NorMuon to standard Muon, leaves the validation BPB essentially unchanged.

6.2 Empirical Verification of NorMuon's Scaling Degeneration

Recall that in Section 3, we theoretically showed that NorMuon's adaptive scaling degenerates to an approximately constant factor on square matrices. In this experiment, we empirically verify this theoretical characterization by removing the adaptive scaling on the Q/K/V/O matrices, in which case NorMuon effectively reduces to standard Muon. In the current NanoChat implementation, the Q/K/V/O projection matrices are optimized as full matrices over all attention heads rather than separately on a per-head basis, and hence are square. We also tested per-head optimization of the Q/K/V matrices, but observed no improvement over joint optimization.

Following the NanoChat setup, the total number of training steps is $7,308$, and validation is performed every 250 steps, with an additional validation run at the end of training. We use validation bits per byte (BPB) as the primary evaluation metric, which measures the average number of bits required to encode each byte of the validation data and is invariant to the tokenizer vocabulary size. Lower BPB indicates better language-modeling performance.

As shown in Figure 1, removing NorMuon's adaptive scaling on the square Q/K/V/O matrices leaves the validation BPB essentially unchanged, with the curves of the two optimizers nearly overlapping. For the training loss, the two curves are virtually indistinguishable, and we therefore report them in Figure 3 in the Appendix. This result is consistent with our theoretical analysis: for square matrices, the corresponding adaptive scaling factors of NorMuon degenerates to an approximately constant scalar, effectively eliminating its intended adaptive effect. Consequently, removing this scaling has negligible impact on optimization performance, providing empirical support for our theoretical characterization.

6.3 DGA-Muon Outperforms NorMuon

We now evaluate the proposed DGA-Muon on NanoChat, using NorMuon as the baseline. We retain the same 24-layer, 1.38B-parameter Transformer architecture as in the ablation study above. For this end-to-end evaluation, we increase the data-to-parameter ratio from 10 to 18, which is also closer to the Chinchilla-style scaling regime (Hoffmann et al., 2022).

Note that DGA-Muon and NorMuon have different optimization characteristics due to their distinct adaptive scaling mechanisms, and therefore require different hyperparameter configurations. We first explain the rationale behind our hyperparameter choices for DGA-Muon. Recall that DGA-Muon constructs its adaptive scaling signal from the raw gradients, whereas NorMuon derives it from the orthogonalized gradients. Orthogonalization provides Muon with favorable isotropic spectral properties, but also removes information encoded in the relative magnitudes of different sin-

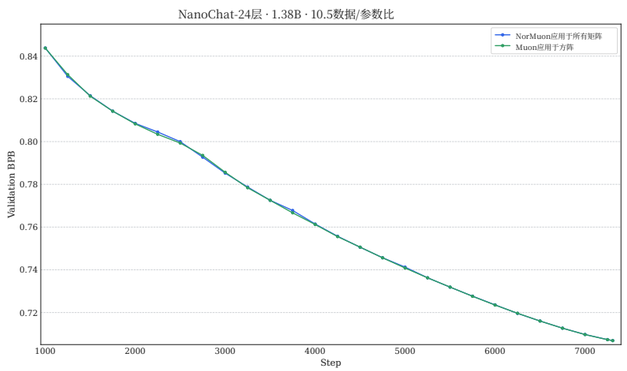


图1：在NanoChat中对方阵上的NorMuon自适应缩放退化进行实证验证。移除方阵Q/K/V/O投影矩阵上的NorMuon自适应缩放，实际上将NorMuon简化为标准Muon，使得验证BPB基本保持不变。

6.2 NorMuon自适应缩放退化实证验证

回想一下，在第 3 节中，我们从理论上证明了 NorMuon 的自适应缩放在方阵上退化为一个近似恒定的因子。在本实验中，我们通过移除 Q/K/V/O 矩阵上的自适应缩放，从经验上验证了这一理论特征，此时 NorMuon 实际上简化为标准 Muon。在当前的 NanoChat 实现中，Q/K/V/O 投影矩阵作为完整矩阵在所有注意力头上进行优化，而不是按头分别优化，因此是方阵。我们还测试了 Q/K/V 矩阵的按头优化，但观察到与联合优化相比没有改进。

遵循 NanoChat 设置，训练步骤总数为 7,308，每 250 步进行一次验证，并在训练结束时进行一次额外的验证。我们使用验证比特每字节（BPB）作为主要评估指标，它衡量编码每个字节验证数据所需的平均比特数，并且不受分词器词汇表大小的影响。较低的 BPB 表示更好的语言模型性能。

如图1所示，移除NorMuon在方形Q/K/V/O矩阵上的自适应缩放因子后，验证BPB基本保持不变，两个优化器的曲线几乎完全重叠。对于训练损失，两条曲线几乎无法区分，因此我们在附录的图3中报告了它们。这一结果与我们的理论分析一致：对于方形矩阵，NorMuon对应的自适应缩放因子退化为一个近似恒定的标量，从而有效地消除了其预期的自适应效果。因此，移除这种缩放对优化性能的影响可以忽略不计，为我们的理论表征提供了经验证据。

6.3 解耦几何对齐自适应缩放公式 优于 NorMuon

我们现在在 NanoChat 上评估所提出的解耦几何对齐自适应缩放公式，使用 NorMuon 作为基线。我们保留了与消融研究相同的 24 层、1.38B 参数 Transformer 架构。为此端到端评估，我们将数据到参数比例从 10 增加到 18，这也更接近 Chinchilla 风格的缩放状态（Hoffmann 等人，2022 年）。

请注意，DGA-Muon 和 NorMuon 由于其不同的自适应缩放机制，具有不同的优化特性，因此需要不同的超参数配置。我们首先解释 DGA-Muon 超参数选择背后的原理。回想一下，DGA-Muon 从原始梯度中构建其自适应缩放信号，而 NorMuon 则从正交化梯度中获取它。正交化为 Muon 提供了有利的各向同性频谱特性，但也去除了不同方向相对幅度中编码的信息。随着训练后期整体更新幅度的减小，更精细的方向差异对于有效优化变得越来越重要，因此这些信息变得日益有价值。相比之下，原始梯度信号保留了这种各向异性信息，但表现出更大的步进间变化。这些特性促使 DGA-Muon 采用更渐进的学习率计划。具体来说，我们延长了预热阶段，增加了冷却率，并降低了最终学习率比例，同时略微降低了矩阵值参数的学习率。完整的超参数配置在附录的表1中提供。

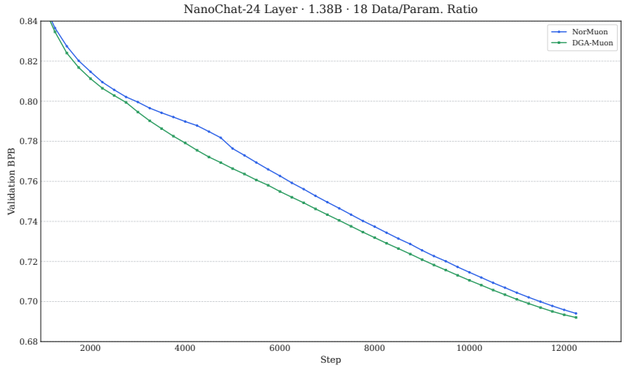


Figure 2: Comparison of DGA-Muon and NorMuon in NanoChat. DGA-Muon consistently outperforms NorMuon, achieving lower validation BPB throughout training.

gular directions. This information becomes increasingly valuable in the later stages of training, as the overall update magnitude decreases and finer-grained differences across singular directions become more important for effective optimization. In contrast, the raw-gradient signal retains this anisotropic information, while exhibiting greater step-to-step variation. These characteristics motivate a more gradual learning-rate schedule for DGA-Muon. Specifically, we extend the warmup phase, increase the warmdown ratio, and reduce the final learning rate fraction, while slightly lowering the learning rate for matrix-valued parameters. The complete hyperparameter configurations are provided in Table 1 in the Appendix.

As shown in Figure 2, DGA-Muon significantly outperforms NorMuon in the end-to-end evaluation, achieving a final validation BPB of 0.6910, compared with 0.6924 for NorMuon. The difference in training loss is more pronounced, with DGA-Muon reaching a final loss of 2.2708 compared with 2.2799 for NorMuon; the corresponding training-loss curves are provided in Figure 4 in the Appendix. We further evaluate the downstream performance of the two optimizers on the DCLM CORE benchmark, the standard aggregate evaluation metric used in NanoChat. DGA-Muon achieves a CORE score of 0.3089, while NorMuon obtains 0.2942, yielding an absolute improvement of 0.0147 and a relative gain of 5.0%. Full downstream results are provided in Table 2 in the Appendix. These results validate the effectiveness of our proposed adaptive-scaling design. By constructing adaptive scaling signals directly from raw gradients and aligning the scaling with the underlying geometry of matrices, DGA-Muon achieves more effective adaptation than NorMuon, leading to lower training loss and validation BPB, as well as better downstream performance.

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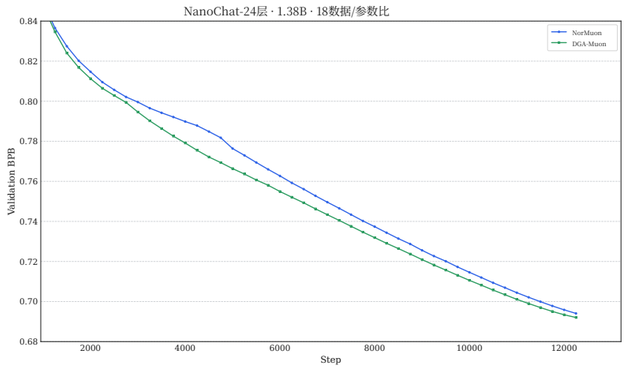


图2：NanoChat中DGA-Muon与NorMuon的比较。DGA-Muon始终优于NorMuon，在整个训练过程中实现了更低的验证BPB。

这些特性促使 DGA-Muon 采用更渐进的学习率计划。具体来说，我们延长了预热阶段，增加了冷却率，并降低了最终学习率比例，同时略微降低了矩阵值参数的学习率。完整的超参数配置在附录的表1中提供。

如图2所示，DGA-Muon在端到端评估中显著优于NorMuon，最终验证BPB达到0.6910，而NorMuon为0.6924。训练损失的差距更为明显，DGA-Muon最终损失为2.2708，而NorMuon为2.2799；相应的训练损失曲线见附录图4。我们进一步评估了这两种优化器在DCLM CORE基准上的下游性能，这是NanoChat中使用的标准综合评估指标。DGA-Muon获得CORE得分为0.3089，NorMuon获得0.2942，绝对提升为0.0147，相对增益为5.0%。完整下游结果见附录表2。这些结果验证了我们提出的自适应缩放设计的有效性。通过直接从原始梯度构建自适应缩放信号，并将缩放与矩阵的底层几何结构对齐，DGA-Muon实现了比NorMuon更有效的自适应，从而降低了训练损失和验证BPB，以及更好的下游性能。

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APPENDIX

A ADDITIONAL EXPERIMENTAL DETAILS

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附录

A 其他实验细节

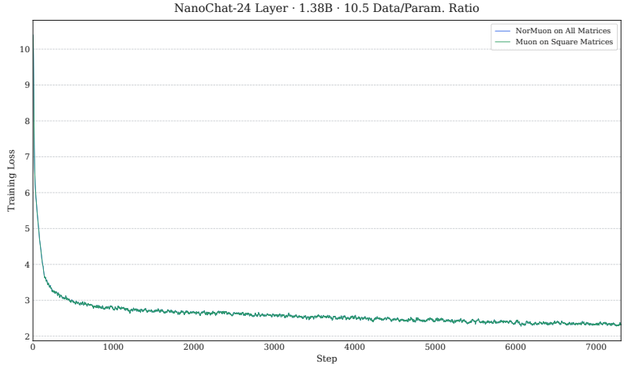


Figure 3: Empirical verification of NorMuon's adaptive scaling degeneration on square matrices in NanoChat. Removing NorMuon's adaptive scaling on the square Q/K/V/O projection matrices, effectively reducing NorMuon to standard Muon, yields virtually indistinguishable training losses.

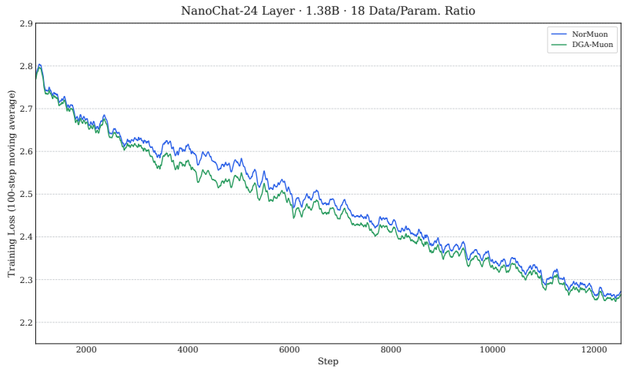


Figure 4: Comparison of DGA-Muon and NorMuon in NanoChat. DGA-Muon consistently outperforms NorMuon, achieving lower training loss throughout training.

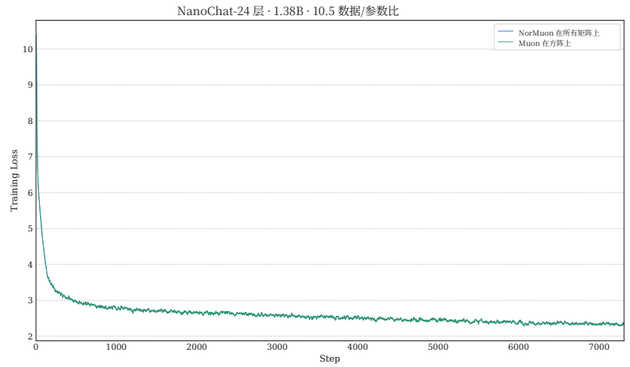


图3：在NanoChat中对NorMuon自适应缩放在方阵上的经验验证。移除NorMuon在方阵Q/K/V/O投影矩阵上的自适应缩放，实际上将NorMuon简化为标准Muon，得到的训练损失几乎无法区分。

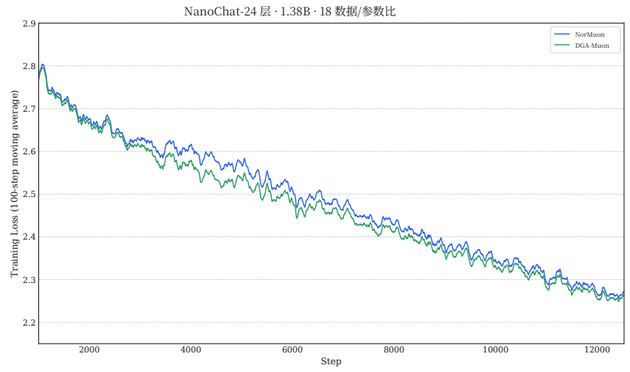


图4：在NanoChat中DGA-Muon与NorMuon的比较。DGA-Muon始终表现优于NorMuon，形成Muon，在整个训练过程中实现更低的训练损失。

Table 1: Hyperparameter configurations for the NorMuon baseline and DGA-Muon in NanoChat.

Hyperparameter	NorMuon Baseline	DGA-Muon
Device batch size	32	32
Total batch size	Auto	Auto
Weight decay	0.28	0.28
Embedding LR	0.3	0.3
Unembedding LR	0.008	0.008
Scalar LR	0.5	0.5
Matrix LR	0.020	0.019
Warmup steps	40	200
Warmdown ratio	0.65	0.78
Final LR fraction	0.05	0.01
Lower scaling clip	–	0.001
Upper scaling clip	–	20

Table 2: Downstream task performance of NorMuon and DGA-Muon on the DCLM CORE benchmark. Higher scores indicate better performance.

Task	NorMuon	DGA-Muon	Diff.
HellaSwag ZeroShot	0.6080	0.6100	**+0.0020**
Jeopardy	0.1680	0.1840	**+0.0160**
BIG-Bench QA Wikidata	0.5300	0.5560	**+0.0260**
ARC-Easy	0.7460	0.7420	−0.0040
ARC-Challenge	0.4760	0.4580	−0.0180
COPA	0.6400	0.7100	**+0.0700**
CommonsenseQA	0.2300	0.2560	**+0.0260**
PIQA	0.7800	0.7840	**+0.0040**
OpenBookQA	0.4320	0.4080	−0.0240
LAMBADA	0.4880	0.4860	−0.0020
HellaSwag	0.6180	0.6260	**+0.0080**
Winograd	0.7143	0.6996	−0.0147
Winogrande	0.5760	0.5660	−0.0100
BIG-Bench Dyck Languages	0.0660	0.0940	**+0.0280**
AGI-Eval LSAT-AR	0.2478	0.3130	**+0.0652**
BIG-Bench CS Algorithms	0.3940	0.4220	**+0.0280**
BIG-Bench Operators	0.2238	0.1857	−0.0381
BIG-Bench Repeat Copy Logic	0.0312	0.0312	0.0000
SQuAD	0.5100	0.4740	−0.0360
CoQA	0.3220	0.3100	−0.0120
BoolQ	0.5660	0.6280	**+0.0620**
BIG-Bench Language Identification	0.2640	0.2500	−0.0140
CORE	**0.2942**	**0.3089**	**+0.0147**

表1：NorMuon基线和DGA-Muon在NanoChat中的超参数配置。

超参数	NorMuon基线	DGA-Muon
设备批大小	32	32
总批大小	Auto	Auto
权重衰减	0.28	0.28
嵌入LR	0.3	0.3
解嵌入学习率	0.008	0.008
标量学习率	0.5	0.5
矩阵学习率	0.020	0.019
预热步数	40	200
冷却率	0.65	0.78
最终学习率分数	0.05	0.01
下限缩放裁剪	–	0.001
上限缩放裁剪	–	20

表2：NorMuon和DGA-Muon在DCLM CORE基准上的下游任务性能。分数越高表示性能越好。

Task	NorMuon	DGA-Muon	Diff.
HellaSwag 零样本	0.6080	0.6100	**+0.0020**
Jeopardy	0.1680	0.1840	**+0.0160**
BIG-Bench QA Wikidata	0.5300	0.5560	**+0.0260**
ARC-Easy	0.7460	0.7420	−0.0040
ARC-Challenge	0.4760	0.4580	−0.0180
COPA	0.6400	0.7100	**+0.0700**
常识问答	0.2300	0.2560	**+0.0260**
PIQA	0.7800	0.7840	**+0.0040**
OpenBookQA	0.4320	0.4080	−0.0240
LAMBADA	0.4880	0.4860	−0.0020
HellaSwag	0.6180	0.6260	**+0.0080**
Winograd	0.7143	0.6996	−0.0147
Winogrande	0.5760	0.5660	−0.0100
BIG-Bench Dyck语言	0.0660	0.0940	**+0.0280**
AGI-Eval LSAT-AR	0.2478	0.3130	**+0.0652**
BIG-Bench 计算机算法	0.3940	0.4220	**+0.0280**
BIG-Bench 操作符	0.2238	0.1857	−0.0381
BIG-Bench 重复复制逻辑	0.0312	0.0312	0.0000
SQuAD	0.5100	0.4740	−0.0360
CoQA	0.3220	0.3100	−0.0120
BoolQ	0.5660	0.6280	**+0.0620**
BIG-Bench 语言识别	0.2640	0.2500	−0.0140
CORE	**0.2942**	**0.3089**	**+0.0147**